

RESEARCH ARTICLE

Measurement Error Estimation and Its Applications in Bearing-Only Localization

BENCAI WANG¹, ZHIYONG WANG¹, HONGTAO ZHAO,
XIAOTONG QI, ZHENG ZHANG, AND LINGYAN ZHAO

Wuhan Electronic Information Institute, Wuhan 430019, China

Corresponding author: Bencai Wang (springbook7559@126.com)

ABSTRACT To clarify the impact of the sensor-target geometry on bearing measurement error estimation, this study conducts an in-depth investigation into the optimal placement of homogeneous/heterogeneous sensors and the optimal bearing measurement error estimation based on bearing-only localization technology. Firstly, a weighted least squares (WLS) bearing measurement error estimation algorithm is proposed, and its influencing factors are analyzed. Secondly, the equivalence among the A-, D-, and E-optimality criteria in the sense of mean squared error (MSE) is obtained, as well as the optimal placement of homogeneous/heterogeneous sensors. Heterogeneous and homogeneous sensors share the same optimal placement form if the bearing measurement accuracy of the sensor is regarded as the weight of distance, i.e., each sensor is equidistant from the target and the separation angles between adjacent sensors are equal. Furthermore, key issues such as the optimal MSE, optimal (estimation, performance, and accuracy) indices, optimal bearing measurement error, and the relationship between MSE and bearing measurement error are discussed. It is pointed out that each bearing measurement error is zero in the statistical sense when the sensors are optimally placed, and the system can achieve the optimal localization accuracy by simultaneously satisfying both the relative sensor-target geometry condition and the measurement error condition. Finally, the specific application of bearing measurement error estimation in the error quadrilateral is presented, and main content and conclusions of this paper are verified by simulations.

INDEX TERMS Bearing measurement accuracy, bearing measurement error estimation, bearing-only localization, error quadrilateral, heterogeneous/homogeneous sensors, optimality criteria, optimal sensor placement.

I. INTRODUCTION

Without loss of generality, the estimation model of passive localization is shown in Fig. 1 [1]. The “localization technology” focuses on the measuring target characteristics acquired by the system, to enable the mathematical determination of its position. For example, if the system obtains angle, time, frequency or signal strength information of the target, it is respectively called the angle of arrival (AOA) or bearing-only (BO), time of arrival (TOA) or time difference of arrival (TDOA), frequency difference of arrival (FDOA), and received signal strength (RSS) localization technology [1], [2], [3], [4], [5], [6]. Without considering the systematic error (hereafter the same), it is necessary to study the correspond-

ing “localization algorithm” to obtain the optimal solution of the target position since the above information contains random measurement error [7], [8], [9]. Different algorithms have differences in terms of unbiasedness, consistency or effectiveness, etc., thus different estimation effects will be produced [5], [10]. This requires examining issues including but not limited to whether the estimation result of algorithm is accurate and under what condition the algorithm performance can be improved, thereby extending into “performance analysis” considerations. This problem mainly consists of three aspects as shown in Fig. 1, the performance analysis of the algorithm is just one of them. The specific analysis proceeds as follows.

- Estimation indices: Analyze the unbiasedness, consistency, or effectiveness of the algorithm, which can

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enhance the localization accuracy through the algorithm performance improvement [11], [12], [13], [14].

- Performance indices: The localization error covariance matrix is analyzed, and the performance indices including mean squared error (MSE), Cramér-Rao lower bound (CRLB), and Fisher information matrix (FIM) are utilized to determine either the optimal trajectory of mobile sensors or the optimal placement of stationary sensors [2], [3], [15], [16] (since the optimal trajectory and the optimal placement are essentially equivalent in nature, the forthcoming sections will not differentiate between sensor mobility states and will uniformly refer to both as “optimal sensor placement.”), and the optimal target position estimation can be achieved.
- Accuracy indices: The localization error is analyzed using accuracy indices including error ellipse (EE) [17], [18], [19], circular error probable (CEP) [20], [21], error quadrilateral (EQ) [18], [22] (following the concept of EE, this paper defines the localization uncertainty region represented by a quadrilateral as the error quadrilateral; see Section IV for details.), and geometric dilution of precision (GDOP) [20], [23], which can obtain the optimal sensor placement with the same objective as previously stated.

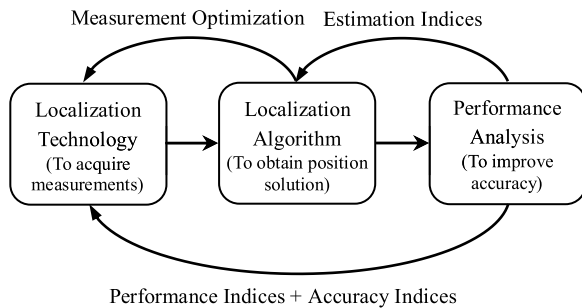


FIGURE 1. Estimation model of passive localization.

In estimation indices, unbiasedness is manifested as the expected value of the target position estimation being the true position value, whereas consistency and effectiveness are shown respectively in the way that the greater the number of measurements or the smaller the variance of the target position estimation, the better the localization accuracy will be. Obviously, each value in the covariance matrix is smaller when the above three indices are better. In the discussion of Section III, it is pointed out that the localization algorithm will simultaneously (approximately) have the above three properties when the sensor is optimally placed relative to the target.

In performance indices, MSE is the trace of the covariance matrix when the target position estimation is unbiased [20], CRLB is the true value or the lower bound of the covariance matrix, and FIM is the inverse matrix of CRLB. In terms of form, it is necessary to transform CRLB and FIM into a scalar form like the MSE through a certain optimality criterion

in practical applications since both of them are matrices. In terms of criterion design, the optimality criteria of the above three kinds of indices are all designed around the minimization of each value in the covariance matrix since the better localization accuracy can be obtained when each value in it is smaller. The A-, D-, and E-optimality criteria are commonly used taking CRLB as an example, which aim to minimize the trace, determinant, and maximum eigenvalue of it respectively [4], [5], [14], [24], [25]. Obviously, the optimality criteria of FIM are exactly opposite to CRLB, which aim to maximize the trace, determinant, and minimum eigenvalue of it respectively [26]. When the sensors are in the optimal placement, MSE is approximately the trace of CRLB. Therefore, A-optimality criterion of CRLB corresponds to that MSE obtains the minimum value [14], [27], and it is equivalent to the D-optimality criterion of FIM under specific conditions [28].

In accuracy indices, EE, CEP, and EQ respectively use an ellipse, a circle, and a quadrilateral to represent the estimation error or the ambiguous area of the target position. Therefore, the better localization accuracy can be obtained when the above areas are smaller. Different from the way that the above three indices use the ambiguous area to measure the localization accuracy, GDOP is a positive proportional function of the covariance matrix trace [20]. Therefore, the first three indices usually aim to minimize the ambiguous area in the optimal sensor placement problem, whereas GDOP aims to minimize its own value. The area of EE is a positive proportional function of the covariance matrix determinant when the measurement error is Gaussian white noise [16], [28], [29], which means that the optimization analysis of EE is equivalent to the D-optimality criteria of performance indices [30]. Similarly, the optimization analysis of CEP and GDOP is equivalent to the A-optimality criterion of performance indices since both of them are positive proportional functions of the covariance matrix trace [20].

In conclusion, all three kinds of aforementioned indices are closely related to the covariance matrix. The performance analysis of estimation indices aims to optimize the localization performance of algorithm among them, whereas the performance and accuracy indices mainly optimize the relative sensor-target geometry. Their common objective is to minimize each value in the covariance matrix, thereby achieving the optimal localization accuracy. The minimization of each element in the covariance matrix is equivalent to its A-optimality criterion, i.e., to minimize the MSE [31], [32]. The localization accuracy is optimal when the sensor placement is optimal with a fixed target, which means each value in the covariance matrix can attain the minimum value. The covariance matrix approximates a scalar one at this time, i.e., its main diagonal elements are equal and the off-diagonal elements are zeros [32], [33]. It should be noted that since the trace of FIM incorporates both the trace and determinant of CRLB, the A-optimality criterion of FIM only strictly opposes the CRLB when it is a scalar matrix. In this case, the components of target position estimation are uncorrelated.

Therefore, as the sensor placement approaches the optimal state, the error covariance matrix converges toward the scalar one. This weakens the correlation between the components of the target position estimation, and consequently, the localization accuracy of the target improves naturally.

As mentioned in [19], [34], [35], [36], and [37], passive localization accuracy depends on not only the relative sensor-target geometry but also the measurement accuracy or measurement error. Combined with Fig. 1 and the above analysis, algorithm performance is the third factor that can affect localization accuracy. The relative sensor-target geometry factor corresponds to the “performance indices” and “accuracy indices” in Fig. 1, whereas the algorithm performance factor corresponds to the “estimation indices”. Measurement accuracy/error factor corresponds to “measurement optimization” in Fig. 1 based on this paper, and it is related to “localization technology” obviously. Based on the main research of this paper, the following discussion primarily focuses on the first two factors.

Many studies have examined the influence of relative geometry factor on localization accuracy based on the performance or accuracy indices in Fig. 1, and pointed out that the localization accuracy will be optimal when the sensors meet the constraint conditions such as being equidistant from the target and having equal separation angles between adjacent sensors; and the sensor placement at this time is referred to optimal [3], [15], [26], [27], [28], [38]. Therefore, the above mentioned constraint conditions are sufficient and necessary for the optimal localization accuracy. Regarding the measurement accuracy/error factor, the better localization accuracy can be obtained when the measurement accuracy of sensor is higher which can lead to smaller measurement error. References [39] and [40] take the measurement errors as the constraint condition which can reflect the underlying sensor-target geometry to obtain the measurement errors of sensors, and then use the difference between the measurement values and the estimation values of measurement errors for target localization. Since the measurement correction values reduce the measurement errors obviously, it can significantly improve the localization accuracy of the target. There is no localization error when the measurement errors are zeros ideally, and the optimal localization accuracy can be obtained without the condition of optimal sensor placement. Therefore, the sensors with no measurement errors and taking the optimal placement have similar effects on the localization accuracy. A core question naturally emerges inspired by above discussion, i.e., does the relative sensor-target geometry has an impact on measurement errors of sensors? In other words, is there a correlation between the other two factors that affect the localization accuracy apart from the algorithm performance?

As previously mentioned, numerous pioneering studies have been conducted regarding the relative sensor-target geometry factor. However, limited research has addressed the optimal placement of heterogeneous sensors (defined in this paper as sensors with varying measurement accuracy) to

our best knowledge. Notably, investigating this issue would substantially enhance the collaborative localization capability of heterogeneous sensors. In terms of measurement accuracy/error factor, there are also few studies addressing the relationship between the relative sensor-target geometry and measurement error. Obviously, the research on this problem will help to clarify the laws of random measurement error, and thus effectively improve the localization accuracy of the target equally with [39] and [40]. As can be seen from Fig. 1, both of these two factors revolve around measurements. Only the former enables the sensors to obtain measurements at the optimal positions, while the latter focuses on reducing measurement errors.

In response to the above problems, this paper first proposes a weighted least squares (WLS) bearing measurement error estimation algorithm based on the BO passive localization technology, and the influencing factors of the algorithm are discussed. In order to clarify the influence of the relative sensor-target geometry factor on the bearing measurement errors, Section III focuses on analyzing the optimal sensor placement of homogeneous/heterogeneous sensors in the sense of MSE and obtains the equivalence among three kinds of optimization criteria. The optimal MSE, indices, bearing measurement error and other issues are discussed simultaneously. In addition to improving the localization accuracy, Section IV points out another important application of the bearing measurement error estimation, i.e., the area of EQ can be obtained, and the relevant factors affecting this area are analyzed. In order to verify the main content of this paper, Section V designs two localization scenarios: fixed sensors for a moving target and moving sensors for a fixed target. Simulations are respectively carried out on four aspects, namely the optimal bearing measurement error estimation of homogeneous/heterogeneous sensors, the bearing measurement error when the sensors are in the optimal placement form, the optimal MSE and EQ area. Section VI finally presents the conclusions of this paper, pointing out the enlightening significance and application value of the measurement error estimation and the optimal placement of heterogeneous sensors, and looking ahead to the issues that are worthy of further research.

According to the above mentioned introduction, the main contributions of this paper are as follows.

- Under the framework of using the bearing measurement accuracy of sensors as sensor-target distance weights, the optimal placement of heterogeneous sensors is proposed. Since this placement form is compatible with the special case of homogeneous sensors, it is more universal in practical applications.
- Each bearing measurement error will have the optimal value when the sensors are in the optimal placement relative to the target, i.e., zero in a statistical sense. This means that when the sensors are in the optimal placement, the system is approximately equivalent to using

the true measurement values for localization, so the optimal accuracy can be obtained.

Notation: Superscript T represents the transpose of a vector or matrix; $\|\cdot\|$ represents the Euclidean norm; $E[\cdot]$ represents the expected value of a variable; $\det(\cdot)$ and $\text{tr}(\cdot)$ represents the matrix determinant and trace respectively; $|\cdot|$ is the absolute value sign.

II. WLS ESTIMATION

Consider the following two-dimensional (2D) localization problem. Suppose in the global coordinate system there are n fixed sensors $S_i = (x_i, y_i)^T$ ($i = 1, 2, \dots, n$), or one moving sensor to find the direction of target at n different locations. The target position is denoted as $\mathbf{X} = (x, y)^T$, the bearing measurement of S_i with respect to $\mathbf{X}$ is $\tilde{\theta}_i$. It has

$$\tilde{\theta}_i = \theta_i + \Delta\theta_i \quad (1)$$

where θ_i is the true bearing, $\Delta\theta_i \sim N(0, \sigma_i^2)$ is the additive i.i.d. zero-mean Gaussian measurement error, then $\mathbf{R} = \text{diag}(\sigma_1^2, \sigma_2^2, \dots, \sigma_n^2)$ is the measurement error covariance matrix. WLS algorithm comprises the following two steps.

Step 1: Combined with (1), the measurement equation of S_i can be expressed by

$$\theta_i = \arctan 2(y - y_i, x - x_i) = \tilde{\theta}_i - \Delta\theta_i \quad (2)$$

which can be linearized by Taylor series about the reference point $\mathbf{T} = (\hat{x}, \hat{y})^T$ and retaining the first two terms, yields [20], [41]

$$\theta_i \approx \hat{\theta}_i(\hat{x}, \hat{y}) - \frac{\hat{y} - y_i}{\hat{d}_i^2} \Delta x + \frac{\hat{x} - x_i}{\hat{d}_i^2} \Delta y \quad (3)$$

where $\hat{\theta}_i$ represents the estimated bearing after substituting $\mathbf{T}$ into (2), $\Delta x = x - \hat{x}$ and $\Delta y = y - \hat{y}$ are the estimated errors of target position, $\hat{d}_i = \|\mathbf{T} - S_i\|$ is the distance from $\mathbf{T}$ to S_i . Substituting (3) into (2), yields the following measurement equation about S_i :

$$-\frac{\hat{y} - y_i}{\hat{d}_i^2} \Delta x + \frac{\hat{x} - x_i}{\hat{d}_i^2} \Delta y \approx \tilde{\theta}_i - \hat{\theta}_i - \Delta\theta_i \quad (4)$$

combining n measurement equations, we have the following BO localization equation:

$$\mathbf{A} \Delta \mathbf{X} \approx \mathbf{Z} - \Delta \boldsymbol{\theta} \quad (5)$$

where

$$\mathbf{A} = \begin{bmatrix} -(\hat{y} - y_1)/\hat{d}_1^2 & -(\hat{y} - y_2)/\hat{d}_2^2 & \dots & -(\hat{y} - y_n)/\hat{d}_n^2 \\ (\hat{x} - x_1)/\hat{d}_1^2 & (\hat{x} - x_2)/\hat{d}_2^2 & \dots & (\hat{x} - x_n)/\hat{d}_n^2 \end{bmatrix}^T$$

is the Jacobian coefficient matrix;

$$\Delta \mathbf{X} = [\Delta x \ \Delta y]^T$$

is the error vector of the target position;

$$\mathbf{Z} = [\tilde{\theta}_1 - \hat{\theta}_1 \ \tilde{\theta}_2 - \hat{\theta}_2 \ \dots \ \tilde{\theta}_n - \hat{\theta}_n]^T$$

is the initial error vector of bearing and

$$\Delta \boldsymbol{\theta} = [\Delta\theta_1 \ \Delta\theta_2 \ \dots \ \Delta\theta_n]^T$$

is the vector of bearing measurement error.

The WLS solution of $\Delta \mathbf{X}$ from (5) is [42]

$$\Delta \hat{\mathbf{X}}_{\text{WLS}} \approx (\mathbf{A}^T \mathbf{R}^{-1} \mathbf{A})^{-1} \mathbf{A}^T \mathbf{R}^{-1} \mathbf{Z} \quad (6)$$

substituting (6) into (5), then the estimated value of bearing measurement error is

$$\Delta \hat{\boldsymbol{\theta}} \approx \mathbf{Z} - \mathbf{A} \Delta \hat{\mathbf{X}}_{\text{WLS}} \quad (7a)$$

namely

$$\Delta \hat{\boldsymbol{\theta}} \approx \tilde{\boldsymbol{\theta}} - (\hat{\boldsymbol{\theta}} + \mathbf{A} \Delta \hat{\mathbf{X}}_{\text{WLS}}) \quad (7b)$$

where $\tilde{\boldsymbol{\theta}} = [\tilde{\theta}_1, \tilde{\theta}_2, \dots, \tilde{\theta}_n]^T$, $\hat{\boldsymbol{\theta}} = [\hat{\theta}_1, \hat{\theta}_2, \dots, \hat{\theta}_n]^T$.

Step 2: After obtaining the first estimation of $\Delta \hat{\boldsymbol{\theta}}$, the approximate true bearings can be obtained using $\tilde{\boldsymbol{\theta}} - \Delta \hat{\boldsymbol{\theta}}$. The new $\mathbf{T}$ can be obtained when the approximate true bearings are applied to the corresponding localization algorithm, then the second estimation of $\Delta \hat{\boldsymbol{\theta}}$ can be obtained by repeatedly executing (3) to (7). So far, WLS bearing measurement error estimation algorithm is completed.

In the above process, the WLS solution of the target position can also be obtained as follows:

$$\hat{\mathbf{X}}_{\text{WLS}} = \mathbf{T} + \Delta \hat{\mathbf{X}}_{\text{WLS}} \quad (8)$$

the above process theoretically can be iteratively looped, i.e., set the detection threshold τ , when

$$\|\Delta \hat{\mathbf{X}}_{\text{WLS}}\| < \tau \quad (9)$$

then the iteration stops [39], and take $\Delta \hat{\boldsymbol{\theta}}$ at this time as the estimated bearing measurement error.

WLS algorithm is used to obtain $\hat{\mathbf{X}}_{\text{WLS}}$ and $\Delta \hat{\boldsymbol{\theta}}$ in the above process, so the method for estimating the bearing measurement error is referred as WLS estimation. It is summarized as follows.

WLS bearing measurement error estimation algorithm	
Step 1	Input: $S_i, \mathbf{X}, \tilde{\theta}_i, \mathbf{T}$
	Output: $\Delta \hat{\boldsymbol{\theta}}$ for the first time
	1: $\hat{\theta}_i = \arctan 2(\hat{y} - y_i, \hat{x} - x_i)$
	2: get $\mathbf{A}, \mathbf{Z}$ using (5)
Step 2	3: $\Delta \hat{\mathbf{X}}_{\text{WLS}} \approx (\mathbf{A}^T \mathbf{R}^{-1} \mathbf{A})^{-1} \mathbf{A}^T \mathbf{R}^{-1} \mathbf{Z}$
	4: $\Delta \hat{\boldsymbol{\theta}} \approx \mathbf{Z} - \mathbf{A} \Delta \hat{\mathbf{X}}_{\text{WLS}}$
	Input: $\Delta \hat{\boldsymbol{\theta}}$ for the first time
	Output: $\Delta \hat{\boldsymbol{\theta}}$ for the second time
Step 2	5: update $\tilde{\theta}_i$ using $\tilde{\boldsymbol{\theta}} - \Delta \hat{\boldsymbol{\theta}}$
	6: update $\mathbf{T}$
	7: repeat 1-4

If $n = 2$ in this method, then $\mathbf{Z} = \mathbf{0}_{n \times 1}$ and makes $\Delta \hat{\boldsymbol{\theta}} = \mathbf{0}_{n \times 1}$. Therefore, the number of measurements needs to

satisfy $\min(n) = 3$ in WLS algorithm. It can be seen from (7) that $\Delta\hat{\theta}$ is related to three factors such as $\mathbf{Z}$, $\mathbf{R}$ and $\mathbf{A}$, which will be discussed separately as follows.

- Combining (3), and (5) to (7), it can be known that $E[\hat{\theta}] \approx \theta = [\theta_1, \theta_2, \dots, \theta_n]^T$ holds if the initial estimate $\mathbf{T}$ of target position is an unbiased estimate of $\mathbf{X}$. It leads to $E[\mathbf{Z}] \approx \mathbf{0}$ and $E[\Delta\hat{\mathbf{X}}_{\text{WLS}}] \approx \mathbf{0}_{2 \times 1}$ in (7a), and makes $\Delta\hat{\theta}$ an approximate unbiased estimate. Therefore, it can be seen that the estimation accuracy of $\Delta\hat{\theta}$ depends on the estimation accuracy of the initial target position from $\mathbf{Z}$ factor. According to (7b) that $\hat{\theta} + \mathbf{A}\Delta\hat{\mathbf{X}}_{\text{WLS}}$ can be regarded as θ , which indicates that $\hat{\theta} \rightarrow \theta$ and $\mathbf{A}\Delta\hat{\mathbf{X}}_{\text{WLS}} \rightarrow \mathbf{0}$ with the better unbiasedness of $\mathbf{T}$. If $\mathbf{T} = \mathbf{X}$ ideally, then $\hat{\theta} = \theta$ holds. But $\Delta\hat{\mathbf{X}}_{\text{WLS}} \neq \mathbf{0}$ from (6) and its value is only $\mathbf{0}$ in a statistical sense, this leads to $\Delta\hat{\theta} \approx \mathbf{Z} = \Delta\theta$. This indicates that an estimation error still exists in the WLS estimation algorithm due to $\Delta\hat{\mathbf{X}}_{\text{WLS}} \approx \mathbf{0}$ even with the best estimation accuracy of $\mathbf{T}$, obviously this is an inherent defect of it.
- Combining (5) to (7), it can be known that $\Delta\hat{\theta} = \mathbf{0}$ for $\mathbf{Z} = \mathbf{0}$ and $\Delta\hat{\mathbf{X}}_{\text{WLS}} = \mathbf{0}$ if $\mathbf{R} = \mathbf{0}_{n \times n}$. This indicates that the estimation accuracy of $\Delta\hat{\theta}$ will be better with the higher measurement accuracy of sensors in terms of $\mathbf{R}$ factor.
- It can be known that $\mathbf{A}$ contains $\hat{d}_i$ and $\hat{\theta}_i$ from (5), so it can reflect the relative geometry between each sensor (or each sensor position, hereinafter referred to as “each sensor.”) and $\hat{\mathbf{X}}_{\text{WLS}}(\mathbf{X})$. Since this relative geometry is an important factor that affects the estimation accuracy of $\Delta\hat{\mathbf{X}}_{\text{WLS}}$ [3], [14], [15], [24], [26], [43], [44], so factor $\mathbf{A}$ will seriously affect the estimation accuracy of $\Delta\hat{\theta}$ according to (7). Thus, the question in the introduction has been answered. Next, we will focus on discussing how to configure the sensors so that the bearing measurement error can be optimally estimated.

III. PERFORMANCE ANALYSIS

Assume that $\mathbf{T}$ is an unbiased estimate of $\mathbf{X}$, then according to (6) the estimation error covariance matrix of $\hat{\mathbf{X}}_{\text{WLS}}$ is

$$\mathbf{P}(\hat{\mathbf{X}}_{\text{WLS}}) = (\mathbf{A}^T \mathbf{R}^{-1} \mathbf{A})^{-1} = \hat{\Phi}^* / \det(\hat{\Phi}) \quad (10)$$

where $\hat{\Phi} = \mathbf{A}^T \mathbf{R}^{-1} \mathbf{A}$, and

$$\hat{\Phi}^* = \begin{bmatrix} \sigma_{\hat{\mathbf{X}}_{\text{WLS}}}^2 & \sigma_{\hat{\mathbf{X}}_{\text{WLS}}\hat{\mathbf{y}}_{\text{WLS}}} \\ \sigma_{\hat{\mathbf{X}}_{\text{WLS}}\hat{\mathbf{y}}_{\text{WLS}}} & \sigma_{\hat{\mathbf{y}}_{\text{WLS}}}^2 \end{bmatrix} \quad (11)$$

is the adjoint matrix of $\hat{\Phi}$, where [20]

$$\sigma_{\hat{\mathbf{X}}_{\text{WLS}}}^2 = \sum_{i=1}^n \left(\cos \hat{\theta}_i / (\hat{d}_i \sigma_i) \right)^2 \quad (12a)$$

$$\sigma_{\hat{\mathbf{y}}_{\text{WLS}}}^2 = \sum_{i=1}^n \left(\sin \hat{\theta}_i / (\hat{d}_i \sigma_i) \right)^2 \quad (12b)$$

$$\sigma_{\hat{\mathbf{X}}_{\text{WLS}}\hat{\mathbf{y}}_{\text{WLS}}} = - \sum_{i=1}^n \sin \hat{\theta}_i \cos \hat{\theta}_i / (\hat{d}_i \sigma_i)^2 \quad (12c)$$

and

$$\det(\hat{\Phi}) = \sum_{i=1}^{n-1} \sum_{j=i+1}^n \left(\sin \Lambda \hat{\theta}_{ij} / (\hat{d}_i \sigma_i \hat{d}_j \sigma_j) \right)^2 \quad (13)$$

where $\Lambda \hat{\theta}_{ij} = \pm (\hat{\theta}_j - \hat{\theta}_i) \in [0, \pi]$ represents the separation angle between S_i and S_j with respect to $\mathbf{T}$. It can be known from (13) that $\hat{\Phi}$ will be singular if all sensors are collinear with the target, therefore the placement of multiple sensors should avoid this situation in practical localization. (when $\Lambda \hat{\theta}_{ij} < -\pi$, $\Lambda \hat{\theta}_{ij} = 2\pi + \Lambda \hat{\theta}_{ij}$; when $\Lambda \hat{\theta}_{ij} > \pi$, $\Lambda \hat{\theta}_{ij} = 2\pi - \Lambda \hat{\theta}_{ij}$.)

Since $\hat{\mathbf{X}}_{\text{WLS}}$ is an unbiased estimate of $\mathbf{X}$, then [45]

$$\mathbf{P} \geq \text{CRLB} = \Phi^{-1} \quad (14)$$

where

$$\Phi = \begin{bmatrix} \sigma_y^2 & -\sigma_{xy} \\ -\sigma_{xy} & \sigma_x^2 \end{bmatrix} \quad (15)$$

is the FIM, where σ_x^2 , σ_y^2 , and σ_{xy} are the results of substituting θ_i and d_i into (12) respectively, and $d_i = \|\mathbf{X} - S_i\|$. It can be known from (14) that each value in $\mathbf{P}$ is smaller, i.e., closer to the corresponding value in the CRLB, the better localization accuracy of $\hat{\mathbf{X}}_{\text{WLS}}$ will be obtained. Ideally, $\mathbf{P} \approx \text{CRLB}$ holds if $\hat{\mathbf{X}}_{\text{WLS}}$ is also an (asymptotic) efficient estimator on the basis of unbiasedness [28]. In order to achieve this condition, the quantities in (12) can be made to reach the minimum values as can be seen from (10), which correspond to the A-optimality criterion; or make (13) reach the maximum value, which corresponds to the D-optimality criterion. The relationships among various criteria are analyzed below.

A. CRITERIA RELATIONSHIP

For the A-optimality criterion of CRLB, since MSE is not less than the sum of the diagonal elements reciprocals of FIM [14], [45], [46], therefore

$$\text{MSE} = \text{tr}(\Phi^{-1}) \geq 1/\sigma_x^2 + 1/\sigma_y^2 \quad (16)$$

the “=” holds in this equation if and only if Φ is a diagonal matrix. This means when

$$\sigma_{xy} = 0 \quad (17)$$

MSE can achieve the minimum value. Apply the Cauchy-Schwarz inequality to (16), then

$$\text{MSE} \geq (1/\sigma_x + 1/\sigma_y)^2 / 2 \quad (18)$$

if and only if when

$$\sigma_x = \sigma_y \quad (19)$$

MSE can reach the minimum value on condition that “=” holds in (18). According to (17) and (19), we have

$$\sum_{i=1}^n \sin 2\theta_i / (d_i \sigma_i)^2 = 0 \quad (20a)$$

$$\sum_{i=1}^n \cos 2\theta_i / (d_i \sigma_i)^2 = 0 \quad (20b)$$

under the conditions of (20), both CRLB and FIM will become scalar matrices.

Transform each element of (15) into

$$\sigma_x^2 = \frac{1}{2} \sum_{i=1}^n (1 + \cos 2\theta_i) / (d_i \sigma_i)^2 \quad (21a)$$

$$\sigma_y^2 = \frac{1}{2} \sum_{i=1}^n (1 - \cos 2\theta_i) / (d_i \sigma_i)^2 \quad (21b)$$

$$\sigma_{xy} = -\frac{1}{2} \sum_{i=1}^n \sin 2\theta_i / (d_i \sigma_i)^2 \quad (21c)$$

then (13) transforms into [15], [28]

$$\det(\Phi) = \frac{1}{4} \left[\sum_{i=1}^n \frac{1}{(d_i \sigma_i)^2} \right]^2 - \frac{1}{4} \left[\sum_{i=1}^n \frac{\sin 2\theta_i}{(d_i \sigma_i)^2} \right]^2 - \frac{1}{4} \left[\sum_{i=1}^n \frac{\cos 2\theta_i}{(d_i \sigma_i)^2} \right]^2 \quad (22)$$

under the condition of (20), the maximum value of this formula is

$$\det(\Phi) = \frac{1}{4} \left[\sum_{i=1}^n 1 / (d_i \sigma_i)^2 \right]^2 \quad (23)$$

According to the definition of MSE, then

$$\text{MSE} = (\sigma_x^2 + \sigma_y^2) / \det(\Phi) \quad (24)$$

substitute (21) and (23) into this equation, yields

$$\text{MSE} = 4 / \sum_{i=1}^n 1 / (d_i \sigma_i)^2 = 2 / \sqrt{\det(\Phi)} \quad (25)$$

the following theorem can be concluded by analyzing this equation.

Theorem 1: In BO passive localization, the A-optimality criterion of CRLB, the D-optimality criterion of FIM, and the E-optimality criterion of either CRLB or FIM are mutually equivalent.

Proof: It is easy to see that the A-optimality criterion of CRLB is equivalent to the D-optimality criterion of FIM from (24) or (25) [28].

MSE reaches its minimum value under the conditions of (17) and (19). Both CRLB and FIM become scalar matrices at this time, which indicates that each of them has two equal eigenvalues and are exactly the main diagonal elements. Obviously, the elements on the main diagonals of the two

matrices are reciprocals of each other. The maximum eigenvalue of CRLB attains the minimum value at this moment, which means the minimum eigenvalue of FIM reaches the maximum value [33]. Therefore, the A-optimality criterion of CRLB is also equivalent to the E-optimality criterion of the CRLB/FIM. ■

Obviously, only under the specific conditions of (17) and (19) do the above optimality criteria possess equivalence. As can be seen from (20), this specific condition is related to three factors such as bearing, distance and measurement accuracy of sensors. However, MSE is only related to the latter two ones under these specific conditions according to (25). Compared with (24), this indicates that the above specific conditions can simplify the expression of MSE. For the simplified MSE, the key issue is to clarify under what conditions the distance and measurement accuracy of each sensor should meet in order to minimize the MSE. The following is an analysis of this problem.

B. CONSTRAINED OPTIMIZATION PROBLEM

Connect the baselines between adjacent sensors in sequence, and a polygon can be formed in the 2D plane. Without loss of generality, the target is located either inside or outside this polygon. Here, adjacent sensors refer to those whose separation angle between them is less than π and are adjacent in spatial position. For convenience, the polygon formed by the sensor baselines will hereafter be briefly referred to as “polygon” in this paper.

Taking the A-optimality criterion of CRLB as an example, it corresponds to the following constrained optimization problem:

$$\arg \max_{\{d_i\}, i=1, \dots, n} \sum_{i=1}^n (d_i \sigma_i)^{-2} \quad (26a)$$

$$\text{s.t.} \begin{cases} \sum_{i=1}^n \sin 2\theta_i / (d_i \sigma_i)^2 = 0 \\ \sum_{i=1}^n \cos 2\theta_i / (d_i \sigma_i)^2 = 0 \end{cases} \quad (26b)$$

to analyze this equation, we have the following theorem.

Theorem 2: When the target is located inside/outside the polygon, MSE can attain the minimum value if and only if the separation angles between adjacent sensors are evenly divided as $2\pi/n$ and the $d\sigma$ values of all sensors are equal. When n and $d\sigma$ are fixed, the optimal MSE is the same whether the target is located inside or outside the polygon. At this moment, the sensors are in the optimal placement relative to the target.

Proof: For the objective function in (26a), it will theoretically reach its maximum value if and only if $d_1 = d_2 = \dots = d_n \approx 0$. However, this situation is obviously not reasonable in practical localization considering localization error or safety factors. Assume the minimum allowable distance between sensor and target is d , then for homogeneous sensors (whether sensors are homogeneous is determined according

to the sensor type or measurement accuracy [33]. Since all sensors in this paper are bearing ones, then homogeneous sensors only refer to sensors with the same bearing measurement accuracy. Similarly, heterogeneous sensors only refer to sensors with different bearing measurement accuracy., if and only if

$$d_1 = d_2 = \dots = d_n = d \quad (27)$$

which means (26a) can achieve its maximum value only when the distances between all sensors and the target reach their minimum values, i.e., all sensors are equidistant from the target. For heterogeneous sensors, σ_i (σ) can be regarded as the weight value of d_i (d). Therefore when

$$d_1\sigma_1 = d_2\sigma_2 = \dots = d_n\sigma_n = d\sigma \quad (28)$$

(26a) reaches its maximum value. This equation indicates that when the product of the distance from each sensor to the target and the corresponding bearing measurement accuracy is equal, MSE can attain its minimum value and the corresponding $\det(\Phi)$ reaches its maximum value. Substitute (28) into (25) yields

$$\min(\text{MSE}) = 4(d\sigma)^2 / n \quad (n \geq 3) \quad (29)$$

Substitute (28) into (26b), then

$$\sum_{i=1}^n \sin 2\theta_i = 0 \quad (30a)$$

$$\sum_{i=1}^n \cos 2\theta_i = 0 \quad (30b)$$

it can be seen that two necessary conditions need to be satisfied when MSE achieves the minimum value in (29). One is the distance (and bearing measurement accuracy) constraint in (28), and the other is the bearing constraint in (30). Regarding the distance constraint, only when the sensors with poor/good bearing measurement accuracy are relatively close to/far from the target can the multi-station system obtain better localization accuracy. As a special case, the optimal localization accuracy can be obtained only when the homogeneous sensors are equidistant from the target. The following will focus on the bearing constraint condition.

With (28), (24) transforms into

$$\text{MSE} = n(d\sigma)^2 / \sum_{i=1}^{n-1} \sum_{j=i+1}^n \sin^2 \Lambda\theta_{ij} \quad (31)$$

where $\Lambda\theta_{ij} = \theta_j - \theta_i$ representing the separation angle between S_i and S_j with respect to X . Two cases are discussed as follows.

1) TARGET LOCATED INSIDE THE POLYGON

Taking $n = 3$ as an example, the relative sensor-target geometry is shown in Fig. 2, where “ $\star$ ” represents the X position, the dashed triangle represents the polygon. The separation angles in Fig. 2 obey the following constraint relationships:

$$\Lambda\theta_{12} + \Lambda\theta_{13} + \Lambda\theta_{23} = 2\pi \quad (32)$$

this conclusion also applies to the case of n sensors, i.e., the sum of the separation angles between all adjacent sensors is 2π . Therefore

$$\Lambda\theta_{12} + \Lambda\theta_{23} + \dots + \Lambda\theta_{(n-1)n} + \Lambda\theta_{n1} = 2\pi \quad (33)$$

combining (31) and (33), the following Lagrange function can be constructed.

$$f(\Lambda\theta, \lambda) = \text{MSE} - \lambda(\Lambda\theta_{12} + \dots + \Lambda\theta_{n1} - 2\pi) \quad (34)$$

where $\Lambda\theta = [\Lambda\theta_{12}, \Lambda\theta_{13}, \dots, \Lambda\theta_{(n-1)n}]^T$, λ is the Lagrange multiplier. If (34) has an extreme value, the necessary condition is

$$\nabla f(\Lambda\theta, \lambda) = 0 \quad (35)$$

where “ ∇ ” is gradient operator. After rearrangement we have

$$\Lambda\theta_{12} = \Lambda\theta_{23} = \dots = \Lambda\theta_{n1} = 2\pi/n \quad (36)$$

it is the same as [3], [15], [26], [27], [28], [38], et al, and MSE can achieve a minimum value when all adjacent separation angles are equal when the target is located inside the polygon. As (36) is the unique solution of (35), this local minimum value is also the global minimum value.

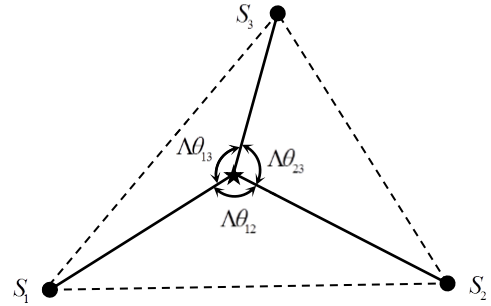


FIGURE 2. Schematic diagram of the target located inside the triangle formed by three sensors.

2) TARGET LOCATED OUTSIDE THE POLYGON

Taking $n = 3$ as an example again, the relative sensor-target geometry is shown in Fig. 3. The separation angles at this point satisfy the following constraints:

$$\Lambda\theta_{12} + \Lambda\theta_{23} = \Lambda\theta_{13} \quad (37)$$

this conclusion also applies to the case of n sensors, i.e., there exists a maximum separation angle at this time which equals to the sum of all other adjacent separation angles. Assuming $\Lambda\theta_{1n}$ is the maximum separation angle for n sensors, then

$$\Lambda\theta_{12} + \Lambda\theta_{23} + \dots + \Lambda\theta_{(n-1)n} = \Lambda\theta_{1n} \quad (38)$$

the remaining analysis process is the same as that of the previous case, and we have

$$\begin{cases} \Lambda\theta_{12} = \Lambda\theta_{23} = \dots = \Lambda\theta_{(n-1)n} = \pi/n \\ \Lambda\theta_{n1} = \sum_{i=1}^{n-1} \Lambda\theta_{i(i+1)} = (n-1)\pi/n \end{cases} \quad (39)$$

same as the conclusion of (36), i.e., MSE can achieve the global minimum value under condition that all the adjacent separation angles are equal when the target is located outside the polygon.

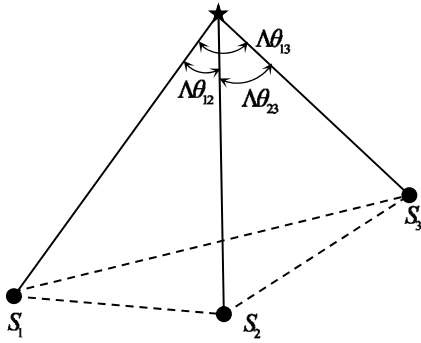


FIGURE 3. Schematic diagram of the target located outside the triangle formed by three sensors.

Another important question need to be solved, i.e., the relationship between (36), (39) and (30). Corollary 3 of [28] takes (36) and (39) as special cases of (30), and this paper will illustrate the equivalence between them by means of graphical representation.

$n \geq 3$ when n is an odd number, take $n = 3$ as an example for illustration. The optimal placement forms corresponding to (36) and (39) are shown in Fig. 4(a) and (b) respectively in special cases, only take Fig. 4(a) as an example for illustration. Assume that the target is located at the origin of the coordinate system, and S_1 is located on the horizontal axis. Since S_2 and S_3 are symmetrical about the horizontal axis, S_1 still remains on the horizontal axis and S_2 and S_3 are also symmetrical about the horizontal axis when θ_1 - θ_3 is doubled. Therefore, (30a) holds for S_1 - S_3 . Similarly, since S'_2 and S'_3 are symmetrical about the vertical axis, (30b) holds for S'_1 - S'_3 . By the same token, (30) also holds since all the sensors except S_1 are symmetric in pairs about the coordinate axes for all odd values of n . In general, the optimal placement forms corresponding to (36) and (39) can be regarded as being obtained by coordinate transformation from Fig. 4(a) and (b) when the target is not located at the origin or S_1 is not on the horizontal axis. Since the bearings of all sensors change by the same amount at this time, (30) still holds.

$n \geq 4$ when n is an even number, take $n = 4$ as an example for illustration. The optimal placement forms corresponding to (36) and (39) are shown in Fig. 4(c) and (d) respectively in special cases. Also assume that the target is located at the origin of the coordinate system, and S_1 is located on the horizontal axis in Fig. 4(c), and all the sensors in (d) are symmetric in pairs about the coordinate axes. Similar to the analysis when n is an odd number, (30) also holds. For all even values of n or in general cases, the analysis process is the same as above.

In conclusion, the separation angle constraint conditions of (36) and (39) are equivalent to the bearing constraint con-

ditions of (30) when the sensors are in the optimal placement relative to the target.

When MSE attains its minimum value, each sensor is in the optimal placement relative to the target since the localization accuracy is optimal. Assume that the separation angle in (36) and (39) is $\Lambda\theta$, then (31) becomes

$$\text{MSE} = n(d\sigma)^2 \left/ \sum_{i=1}^{n-1} (n-i) \sin^2(i\Lambda\theta) \right. \quad (40)$$

then regardless of whether the target is located inside or outside the polygon, MSE obtained from this formula is also the same as that in (29) (see Appendix A for details). This indicates that when the sensors are in the optimal placement relative to the target: (1) When $d\sigma$ and n are fixed values, MSE is the same when the target is located inside and outside the polygon; (2) MSE is only related to two factors of $d\sigma$ and n , and the better MSE will be obtained when $d\sigma$ is smaller or n is larger. Combining (28), (36) and (39), the first conclusion of Theorem 2 can be obtained; from the analysis of (40), the second conclusion of Theorem 2 can be obtained. ■

Obviously, the optimal placement is that each sensor is located at the vertex of a regular polygon centered on the target position when the target is located inside the polygon formed by the homogeneous sensors.

Based on the above mentioned analysis, further discussions will be made below on issues such as constraint conditions, the optimal indices, the optimal estimation of bearing measurement error, et al.

C. DISCUSSION

1) OPTIMAL SEPARATION ANGLE

According to (36) and (39), the adjacent separation angles all approach zero when n approaches infinity. The sum of the separation angles in (36) is always 2π regardless of the value of n , whereas the sum of the separation angles in (39) needs to satisfy $\Lambda\theta_{n1} \leq \pi$. This indicates that the maximum value of the sum of adjacent separation angles is π when the target is located outside the polygon. This result is in line with the actual situation, for it contradicts the condition that the target is located outside the polygon when the sum of the separation angles exceeds π .

2) CONSTRAINT CONDITIONS

Both bearing and distance constraint conditions need to be satisfied when MSE reaches its minimum value. Therefore, the bearing and distance are two important factors affecting the localization accuracy for BO passive localization. The constraint conditions for the separation angles when MSE reaches its minimum value are the same when the target is located inside or outside the polygon, i.e., the adjacent separation angles are equal; the difference lies in that the former evenly divides 2π whereas the latter evenly divides π . The distance constraint conditions are also the same for homogeneous sensors, i.e., each sensor is equidistant from the target. However, the distance constraint conditions require

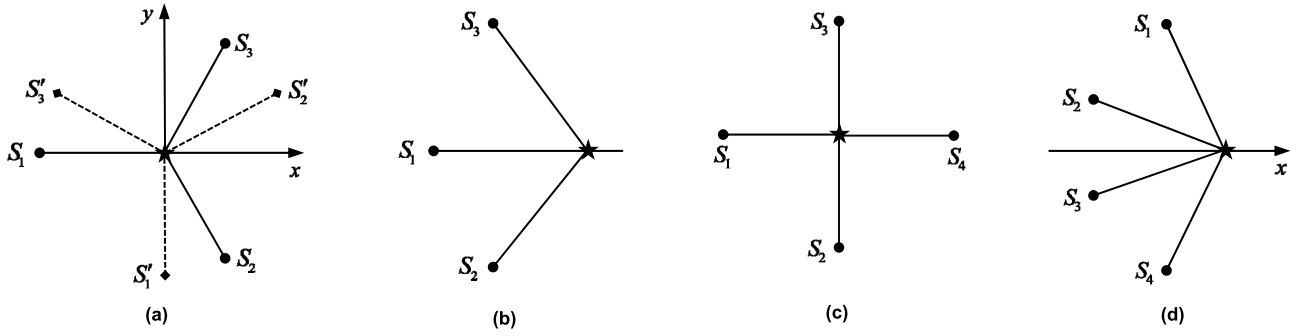


FIGURE 4. Optimal placement form. (a) Target is located inside the triangle formed by three sensors. (b) Target is located outside the triangle formed by three sensors. (c) Target is located inside the quadrilateral formed by four sensors. (d) Target is located outside the quadrilateral formed by four sensors.

that each sensor has the same value of $d\sigma$ for heterogeneous sensors. In other words, the distance constraint conditions for heterogeneous sensors are the same as homogeneous sensors if the bearing measurement accuracy is taken as the weight of the sensor distance. Thus it can be seen that the optimal placement of homogeneous sensors is just a special case of heterogeneous sensors. It should be noted that the bearing constraint conditions in (36) and (39) are also applicable to the optimal placement of other active/passive localization technologies such as TOA or distance measurement [15].

3) OPTIMAL MSE

Combining (16) and (24) we have

$$\max(\det(\Phi)) = \sigma_x^2 \sigma_y^2 \quad (41)$$

substitute (20b) into (21a) and (21b) respectively, yields

$$\sigma_x^2 = \sigma_y^2 = \frac{1}{2} \sum_{i=1}^n \frac{1}{(d_i \sigma_i)^2} \quad (42)$$

substitute this into (41), and (23) can also be obtained. Substitute (28) into (23) and (42) respectively, yields

$$\max(\det(\Phi)) = n^2 / (4 (d\sigma)^4) \quad (43)$$

$$\sigma_x^2 = \sigma_y^2 = n / (2 (d\sigma)^2) \quad (44)$$

substitute the above two equations into (24), (29) can also be obtained. This indicates that the optimal MSE of D-optimality criterion of FIM is the same as the A-optimality criterion of CRLB. It can be seen from (29) that when each d is smaller, the better localization accuracy will be obtained when n is fixed. However, reducing the distance will be limited by practical factors such as sensor safety as mentioned before. The localization accuracy will be better when increasing n on condition that each d is fixed. However, it is obvious that the increase of n leads to a less significant improvement in localization accuracy. Therefore, the balance between the localization cost and accuracy should be considered in practical applications. It should be noted that when the sensors are in a general placement, the homogeneous and heterogeneous sensors will have the same MSE from (24) as long as the

heterogeneous sensor has the same $d\sigma$ and bearing with the corresponding homogeneous sensor. When the sensors are in the optimal placement from the analysis of (40), the multi-station system will obtain the same MSE as long as the heterogeneous sensor has the same $d\sigma$ with the corresponding homogeneous sensor regardless of whether the target is located inside/outside the polygon or whether the sensors belong to homo-/heterogeneous.

4) OPTIMAL INDICES

Here, we discuss the variation of the three types of indices in Fig. 1 when the sensors are in the optimal placement relative to the target. $E[\Delta \hat{\mathbf{X}}_{\text{WLS}}] \approx \mathbf{0}$ holds for the estimation indices since $\mathbf{T} \approx \mathbf{X}$ and $E[\mathbf{Z}] \approx E[\Delta \theta] = \mathbf{0}$ at this time, which means the localization algorithm is unbiased approximately. It can be known from (29) that the localization algorithm is consistent. As mentioned before, the localization algorithm is effective since each element in $\mathbf{P}$ has the minimum value. In summary, the localization algorithm simultaneously (approximately) satisfies the three properties of the estimation indices when the sensors are in the optimal placement form. For the performance indices, either CRLB or FIM is a scalar matrix at this time and the trace, determinant, and maximum/minimum eigenvalue of the CRLB/FIM respectively reaches the minimum/maximum value. Conversely it is also true, i.e., only when $\mathbf{P}$ approaches a scalar matrix can each sensor present the optimal placement relative to the target under the influence of bearing measurement error [33]. For the accuracy indices, the optimal EE generated by (8) will be approximately circular since $\mathbf{P}$ has two approximately equal eigenvalues at this time when the multi-station system obtains the optimal localization accuracy [20], i.e., EE degenerates into an error circle and its area has the minimum value among all EEs. Similarly, both CEP and GDOP reach the minimum value. The area of EQ is a function of bearing measurement error from the subsequent Section IV. Combining with the subsequent part of “optimal estimation of bearing measurement error”, the area of EQ also has the minimum value when the sensors are in the optimal placement form.

5) OPTIMAL ESTIMATION OF BEARING MEASUREMENT ERROR

As what is discussed on the factor $\mathbf{Z}$ in Section II, $\Delta\hat{\mathbf{X}}_{\text{WLS}} \approx \mathbf{0}$ holds when the sensors are in the optimal placement, which in turn makes

$$\Delta\hat{\theta} \approx \mathbf{0} \quad (45)$$

this indicates that the estimation error of the target position approaches $\mathbf{0}$ when the sensors are in the optimal placement, which in turn makes the bearing measurement error of each sensor zero in a statistical sense. This means that in BO localization, the relative sensor-target geometry has a significant impact on the bearing measurement error. One manifestation is that under the assumption that $\Delta\theta$ is Gaussian white noise, $\Delta\hat{\theta}$ can achieve the optimal estimation if and only if the sensors are in the optimal placement form. Just as the constraints of bearing and distance are necessary conditions for the optimal MSE, they are also the necessary conditions for the sensor to obtain the optimal bearing measurement error. Obviously, the optimal bearing measurement error depends only on the estimation accuracy of $\mathbf{T}$. Since the sensors are in the optimal placement at this time, the estimation accuracy of $\mathbf{T}$ can be guaranteed at this time from the discussion of “optimal indices”. The following needs to be emphasized for the optimal bearing measurement error: (i) Step 2 of WLS estimation algorithm fails since $E[\Delta\hat{\theta}] \approx \mathbf{0}$ at this time, i.e., step 2 cannot further improve the estimation accuracy of $\mathbf{T}$ at this moment. (ii) It can be known from (29) that reducing d can improve the estimation accuracy of $\mathbf{T}$ at this time. However, $\Delta\hat{\theta}$ is always $\mathbf{0}$ in a statistical sense regardless of the value of d , which in turn makes $E[\Delta\hat{\theta}]$ a fixed value approximately. This indicates that reducing d cannot effectively improve the estimation accuracy of $\Delta\hat{\theta}$ after the sensors are in the optimal placement. (iii) Just as stated in the “optimal indices”, increasing n can further improve the estimation accuracy of $\Delta\hat{\theta}$ since the WLS algorithm satisfies the consistency of estimation at this time. However, due to the correlation between the coefficient matrix $\mathbf{A}$ and the bearing error, there still exists an estimation bias even when $n \rightarrow \infty$ for $\Delta\hat{\theta}$ is obtained by WLS algorithm [10].

6) THE RELATIONSHIP BETWEEN MSE AND THE BEARING MEASUREMENT ERROR

Since $\mathbf{P}$ is included in (7), it can be known that $\Delta\hat{\theta}$ is closely related to MSE. It can be known that $\Delta\hat{\theta}$ will approach to the true value $\Delta\theta$ if $\Delta\hat{\mathbf{X}}_{\text{WLS}}$ is smaller combining with the discussion of factor $\mathbf{Z}$ in Section II, which means that the better localization accuracy will be obtained with the smaller MSE. If $\mathbf{T} = \mathbf{X}$ ideally, then MSE obtains the minimum value and the localization accuracy is optimal for $\Delta\hat{\mathbf{X}}_{\text{WLS}} \approx \mathbf{0}$ and $\Delta\hat{\theta} \approx \Delta\theta$. It can be seen that the general placement of sensors can also affect the bearing measurement error, one manifestation is that the MSE is directly proportional to the estimation accuracy of $\Delta\hat{\theta}$. Specifically, MSE will be smaller when $\Delta\hat{\theta}$ is closer to $\Delta\theta$, MSE obtains the minimum

value when $\Delta\hat{\theta}$ is approximately equal to $\Delta\theta$. Conversely, MSE will be larger when $\Delta\hat{\theta}$ deviates from $\Delta\theta$. As discussed in the “optimal estimation of bearing measurement error” section, the above relationship no longer holds when the sensors are in the optimal placement form. The bearing and distance constraint conditions in (36), (39) and (27), (28) are the necessary and sufficient conditions for the minimum MSE and the optimal $\Delta\hat{\theta}$ at this time, and the localization system can obtain the optimal localization accuracy since it satisfies both the relative sensor-target geometry condition and the measurement error condition described in the introduction part.

Obviously, bearing measurement error estimation is mainly applied in the algorithm for improving the localization accuracy [39], [40]. As can be seen from (1), an approximate true value of bearing can be obtained by subtracting the estimated bearing measurement error from the bearing measurement value. Therefore, the measurement error estimation is an optimization process of the measurement data based on the localization algorithm combined with Fig. 1. In addition, the estimation of bearing measurement error also has substantial application value in EQ.

IV. AREA OF EQ

Suppose the baseline S_1S_2 is parallel to the horizontal axis of the coordinate system in Fig. 5, the bearings measured by two sensors about target $\mathbf{X}$ are $\tilde{\theta}_1$ and $\tilde{\theta}_2$ respectively. Then two bearing lines can be determined by $\tilde{\theta}_1$ and $\tilde{\theta}_2$ respectively and the intersection point $\mathbf{T}$ is the estimated value of $\mathbf{X}$, which is the principle of direction-finding cross localization [32].

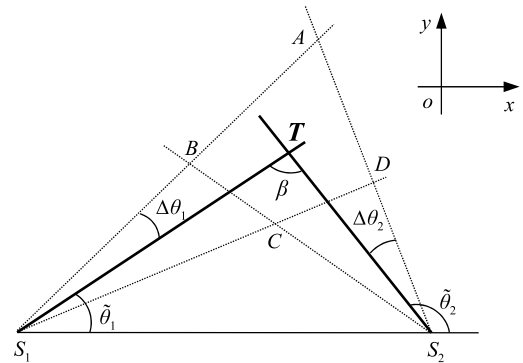


FIGURE 5. Schematic diagram of EQ.

If the bearing measurement errors of two sensors are $\Delta\theta_1$ and $\Delta\theta_2$ respectively, then $\mathbf{X}$ must lie on the ray starting from S_1 with an bearing of $\tilde{\theta}_1 - \Delta\theta_1$ for bearing line $S_1\mathbf{T}$. Due to the randomness of $\Delta\theta_1$, $\mathbf{X}$ must be on S_1D when $\Delta\theta_1 > 0$, and on S_1A when $\Delta\theta_1 < 0$. Similarly for the bearing line $S_2\mathbf{T}$, $\mathbf{X}$ must be on either S_2B or S_2A . Therefore, $\mathbf{X}$ will eventually be at one of the points A , B , C , or D . If $\Delta\theta_1$ or $\Delta\theta_2$ is the maximum value of bearing measurement error of S_1 or S_2 respectively, then $\mathbf{X}$ must be within the area enclosed by $ABCD$. Referring to the concept of EE, the quadrilateral formed by $ABCD$ is called EQ. As stated in introduction, the

better localization accuracy will be obtained when the area of EQ is smaller.

Let the area of EQ be S_{EQ} , from Fig. 4 we have

$$S_{EQ} = S_{\Delta S_2AB} - S_{\Delta S_2DC} \quad (46)$$

where $S_{\Delta S_2AB}$, $S_{\Delta S_2DC}$ represents the area of ΔS_2AB and ΔS_2DC respectively, and

$$S_{\Delta S_2AB} = S_2A \cdot S_2B \cdot \sin(2|\Delta\theta_2|)/2 \quad (47a)$$

$$S_{\Delta S_2DC} = S_2D \cdot S_2C \cdot \sin(2|\Delta\theta_2|)/2 \quad (47b)$$

let $S_1S_2 = L$ and $\beta = \tilde{\theta}_2 - \tilde{\theta}_1$ (when T is below S_1S_2 , then $\beta = \tilde{\theta}_1 - \tilde{\theta}_2$), we have

$$S_2A = \frac{L \sin(\tilde{\theta}_1 + |\Delta\theta_1|)}{\sin(\beta - |\Delta\theta_1| - |\Delta\theta_2|)} \quad (48a)$$

$$S_2B = \frac{L \sin(\tilde{\theta}_1 + |\Delta\theta_1|)}{\sin(\beta - |\Delta\theta_1| + |\Delta\theta_2|)} \quad (48b)$$

$$S_2D = \frac{L \sin(\tilde{\theta}_1 - |\Delta\theta_1|)}{\sin(\beta + |\Delta\theta_1| - |\Delta\theta_2|)} \quad (48c)$$

$$S_2C = \frac{L \sin(\tilde{\theta}_1 - |\Delta\theta_1|)}{\sin(\beta + |\Delta\theta_1| + |\Delta\theta_2|)} \quad (48d)$$

substitute (48) and (47) into (46) and assume that $\Delta\theta_1$ and $\Delta\theta_2$ are small enough, after rearrangement we have (see Appendix B for details)

$$S_{EQ} \approx 4L^2 |\Delta\theta_1 \Delta\theta_2| \sin \tilde{\theta}_1 \sin \tilde{\theta}_2 / \sin^3 \beta \quad (49)$$

this is the EQ area generated by dual-sensor in the local coordinate system. As shown, S_{EQ} depends on three factors such as bearing (separation angle), baseline length, and bearing measurement error. Each factor is analyzed separately below.

- Bearing (Separation angle). It can be known from (49) that it makes sense if and only if

$$\sin \tilde{\theta}_1 \neq 0(\pi) \quad (50a)$$

$$\sin \tilde{\theta}_2 \neq 0(\pi) \quad (50b)$$

$$\sin \beta \neq 0(\pi) \quad (50c)$$

As shown in Fig. 5, the first two formulas require that S_1T/S_2T cannot be parallel to the horizontal axis of the coordinate system and (50c) requires that S_1 , S_2 and T cannot be collinear, which is consistent with the analysis of (13). Generally speaking, two sensors can localize the target as long as the separation angle satisfies (50c) if S_1T or S_2T is parallel to the coordinate axis. However, S_{EQ} cannot be used to measure the localization accuracy at this moment, which is an inherent defect of EQ. In addition, (49) is a quadratic function of bearing but a cubic function of the separation angle, which indicates that the latter will play a vital role in affecting S_{EQ} . For dual-sensor BO system, only when the separation angle

satisfies a specific range/value can the dual-sensor system achieve better/optimal localization accuracy [15].

- Baseline length. Obviously, L is a function of bearing. When the positions of target and one sensor are fixed, S_{EQ} will be larger when L is smaller for β will be smaller. When β is fixed and satisfies the above mentioned specific range/value, S_{EQ} will be smaller when L is smaller for the case of smaller sensor-target distances.
- Bearing measurement error. EE area is an expression related to the bearing measurement accuracy [17], [20], whereas EQ area is an expression about bearing measurement error according to (49). For bearing measurement sensor, its measurement accuracy is fixed but the measurement error is a random variable. Under the combined influence of the internal and external noises of the dual-sensor system, it is generally believed that the bearing measurement error follows a Gaussian distribution with zero-mean and the bearing measurement errors of two sensors are independent of each other. Therefore, the values of $\Delta\theta_1$ and $\Delta\theta_2$ cannot be given in advance when using EQ to measure the localization accuracy, because the estimated values are needed firstly. Compared with the other three accuracy indices such as EE, CEP and GDOP, this indicates that it is relatively cumbersome to use EQ to measure the localization accuracy. But as shown in Fig. 5, as one of the accuracy indices, the greatest advantage of EQ is that it can vividly and intuitively reflect the localization error.

Other discussions about EQ are as follows: (i) EE and CEP become spheres in a three dimensional (3D) localization scenario, whereas EQ will become an irregular and complex polyhedral structure due to the combined influence of bearing and elevation errors. (49) is obtained under the condition that $\Delta\theta_1$ and $\Delta\theta_2$ are small enough, which will lead to an accuracy decrease about it in the situation of large bearing measurement error. Therefore, the application of EQ will be limited in 3D localization or large bearing measurement error scenario. (ii) Obviously, the optimal localization accuracy can be obtained when dual-sensor system is in the optimal placement. As known from the discussion part in the previous section, dual-sensor has the optimal bearing error at this time. As known from the introduction part, EE, CEP and GDOP are all closely related to the covariance matrix. As one of the accuracy index, is EQ also related to it? Intuitively, the answer should be yes. Since this problem falls outside the scope of this study, it will be a promising direction for future research.

V. SIMULATION VERIFICATION

This section mainly focuses on the WLS bearing measurement error algorithm, the constraint conditions for homogeneous/heterogeneous sensors to obtain the optimal placement and relevant issues of EQ. The simulations of the first two aspects are shown in Table 1, and EQ mainly conducts a simulation of (49) when the target is traversed and compares it with the true value. RMS in Table 1 refers to the root mean

TABLE 1. Simulation content of the WLS algorithm and the optimal placement.

Simulation Content	Homogeneous Sensor		Heterogeneous Sensor	
	Inside	Outside	Inside	Outside
Optimal Estimation of Bearing Measurement Error	✓			✓
Bearing Measurement Error When Sensors Take Optimal Placement		✓	✓	
RMS And the Optimal MSE		✓		✓

TABLE 2. Bearing measurement accuracy of heterogeneous sensor.

Sensor	S_1^*	S_2^*	S_3^*	S_4^*	S_5^*	S_6^*	S_7^*
Scenario 1	Bearing	0.7	0.3	0.4	0.3	0.3	0.7
Scenario 2	Measurement Accuracy/ $^\circ$	1.5	0.8	2	0.5	0.7	

square error, i.e., $\text{RMS}^2 = E[\|\hat{\mathbf{X}} - \mathbf{X}\|_2^2]$. “Inside” and “outside” respectively refers to the situation where the target is located inside or outside the polygon. The following two simulation scenarios are designed:

(1) Localizing a moving target with fixed sensors. S_i ($i = 1, \dots, 9$) is homogeneous sensor as shown in Fig. 6, and the bearing measurement accuracy is $\sigma = 0.5^\circ$. S_1 - S_4 and S_5 - S_9 are respectively the vertices of a regular quadrilateral and pentagon, and the distance between each sensor and the origin is $50\sqrt{2}$ km. “T1-T4” represent 4 targets respectively, where the horizontal axis movement range of T1 and T2 is $[-10, 10]$ km. T1 has a straight-line track with a slope of $\tan \pi/6$, T2 has a curved track and its equation is $0.1x^2$. Obviously, S_1 - S_4 and S_5 - S_9 are respectively in the optimal placement when T1 and T2 move to the origin. The tracks of T3 and T4 are the results of the translation of T1 and T2 respectively, where T3 is respectively shifted 20 km to the left and upward relative to T1, and T4 is shifted 30 km to the right relative to T2. S_i^* is a heterogeneous sensor corresponding to S_i , and its bearing measurement accuracy is shown in Table 2. When the target is at the origin, S_1^* , S_2^* , S_3^* , S_4^* and S_5^* , S_6^* , S_7^* , S_8^* , S_9^* are all in the optimal placement since the $d\sigma$ of each sensor is $50\sqrt{2} \times 0.5$. The sensors used in the simulation process are shown in Table 3.

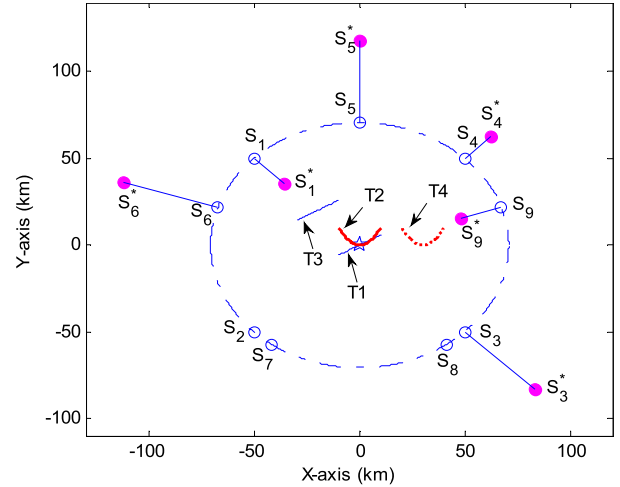
(2) Localizing a fixed target with moving sensors. In Fig. 7, the bearing measurement accuracy of homogeneous sensor S_i ($i = 1, \dots, 6$) is $\sigma = 1^\circ$. Except for S_2 , the remaining sensors are all located on a circle with the target position “ $\star$ ” as the center and a radius of 20 km. S_2 is a moving sensor, where Track 1 moves along the vertical axis and the motion range is $[-30, -20]$ km, Track 2 is a straight line parallel to the horizontal axis and the motion range is $[-10, 10]$ km. S_1 - S_3 , S_2 and S_4 - S_6 are respectively in the optimal placement when S_2 moves to the position shown as “ $\circ$ ” in the figure. Obviously, the bearing between Track 1 and the target remains unchanged, whereas both the bearing and the distance of Track 2 change. S_i^* is also a heterogeneous sensor corresponding to S_i , and its bearing measurement accuracy is shown in Table 2. The $d\sigma$ value of each sensor is 20 when S_2 moves to the “ $\circ$ ” position, so S_1^* , S_2^* , S_3^* and S_4^* , S_5^* , S_6^* are all in the optimal placement. The sensors used in the simulation process are shown in Table 3.

TABLE 3. Usage of sensors.

Sensor		$n = 4/3$	$n = 5/4$
Scenario 1	homogeneous	$S_1 - S_4$	$S_5 - S_9$
	heterogeneous	$S_1^* \ S_2^* \ S_3^* \ S_4^*$	$S_5^* \ S_6^* \ S_7^* \ S_8^* \ S_9^*$
Scenario 2	homogeneous	$S_1 - S_3$	$S_2 \ S_4 - S_6$
	heterogeneous	$S_1^* \ S_2^* \ S_3^*$	$S_4^* \ S_5^* \ S_2 \ S_6^*$

S_i^* are all in the optimal placement. The sensors used in the simulation process are shown in Table 3.

In the WLS algorithm, the TLS algorithm is applied to obtain the initial value of target position $\mathbf{T}$ in Section C [7], and the LS algorithm is applied in the remaining sections. Moreover, only a two-step method is used to obtain the bearing measurement error of each sensor. The Monte Carlo simulation is carried out 5000 times in Section D, and 10000 times in the remaining sections.

**FIGURE 6.** Localization scenario 1.

A. OPTIMAL ESTIMATION OF BEARING MEASUREMENT ERROR

This section mainly verifies that the WLS bearing measurement error estimation algorithm will achieve the optimal estimated value when the sensors are in the optimal placement form.

1) HOMOGENEOUS SENSOR

When using the sensors in two different quantities listed in Table 3 to localize 4 moving targets respectively, the WLS

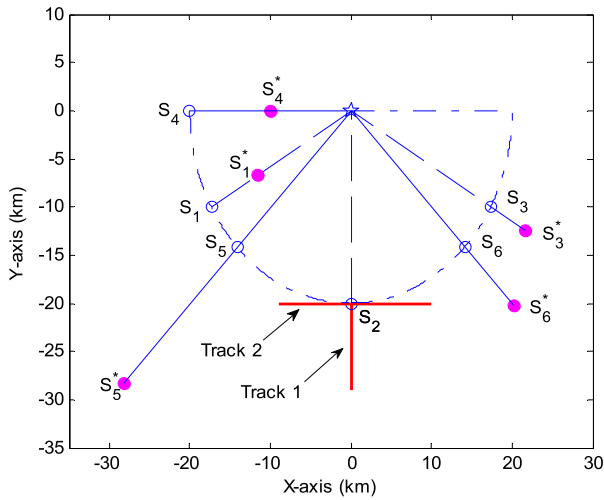


FIGURE 7. Localization scenario 2.

algorithm is employed to estimate the bearing measurement error of each sensor. The sampling interval of each sensor for the target is 1 km, and the results are shown in Fig. 8 and 9. It should be noted that since the bearing measurement error is a random variable, directly taking the mean value of it will cause some values to cancel each other out. Therefore, $E[|\Delta\hat{\theta}|]$ is used to measure the variation of the estimation result of the bearing measurement error, i.e., the absolute value of the estimated value is taken first, and then the mean value is calculated. As can be seen from the figures:

(i) By comparing (a) and (b) in Fig. 8 and 9, $|\Delta\hat{\theta}|$ is relatively concentrated when the sensors are in the optimal placement, whereas this phenomenon does not exist in Fig. (c) and (d). This indicates that when the sensors are in the optimal placement relative to the target, $|\Delta\hat{\theta}_i|$ of each sensor tends to be a fixed value. Each quantity of $\Delta\hat{\theta}$ is 0 at this time in the statistical sense according to the analysis of (45), which lead to each $|\Delta\hat{\theta}_i|$ is approximately equal. This phenomenon first verifies the correctness of (27) and (30), i.e., only when homogeneous sensors are equidistant from the target and the separation angles between adjacent sensors are equal can the optimal placement be obtained; second, the bearing measurement error will achieve the optimal estimation when homogeneous sensors present the optimal placement. At this time, $\Delta\hat{\theta} \approx \Delta\theta$ holds. The fixed values in Fig. 8 and 9 are approximately 0.28° and 0.31° respectively, whereas the absolute value of the true error value is $E[|\Delta\theta|] \approx 0.4^\circ$. This indicates that the WLS bearing measurement error algorithm has the consistency of estimation, i.e., the better estimation accuracy can be obtained when more sensors are used.

(ii) There is a direct proportional relationship between $|\Delta\hat{\theta}_i|$ and d_i as can be seen from (a) and (b) in Fig. 8 and 9, which indicates that the absolute value of the estimation error $E[|\Delta\hat{\theta}_i - \Delta\theta_i|]$ of $\Delta\hat{\theta}_i$ is inversely proportional to d_i . Specifically, $|\Delta\hat{\theta}_i|$ gradually decreases/increases when S_i gets closer

to/farther away from the target, but $|\Delta\hat{\theta}_i - \Delta\theta_i|$ gradually increases/decreases. For example, as the target gets farther away from S_2 in Fig. 8(a), $|\Delta\hat{\theta}_2|$ and $|\Delta\hat{\theta}_2 - \Delta\theta_2|$ becomes larger and smaller respectively. Since d_2 and d_4 change in opposite trends, then $|\Delta\hat{\theta}_2|$ and $|\Delta\hat{\theta}_4|$, $|\Delta\hat{\theta}_2 - \Delta\theta_2|$ and $|\Delta\hat{\theta}_4 - \Delta\theta_4|$ also approximately change in opposite trends respectively. The same phenomenon exists between S_1 and S_3 , i.e., the absolute value of each bearing measurement error and corresponding estimated error also approximately change in opposite trends when d_1 and d_3 change in opposite trends. But compared with S_2 and S_4 , the change range of the absolute values of each bearing measurement error and corresponding estimation error of them also tend to be gentle since the distance changes are not obvious at this time. It can be seen from the other figures that the above phenomenon is universal, i.e., the absolute values of bearing measurement error and corresponding estimation error of each homogeneous sensor are symmetrically distributed when the sensors are in a general placement form, and the center is the fixed value in the optimal placement form. For example, $|\Delta\hat{\theta}_1|$ and $|\Delta\hat{\theta}_2|$ are symmetrical to $|\Delta\hat{\theta}_3|$ and $|\Delta\hat{\theta}_4|$ respectively in Fig. 8. In Fig. 9, $|\Delta\hat{\theta}_5|$ and $|\Delta\hat{\theta}_7|$ are symmetrical to $|\Delta\hat{\theta}_6|$ and $|\Delta\hat{\theta}_9|$ respectively in (a), $|\Delta\hat{\theta}_5|$ is symmetrical to $|\Delta\hat{\theta}_7|$, $|\Delta\hat{\theta}_6|$ is symmetrical to the sum of $|\Delta\hat{\theta}_8|$ and $|\Delta\hat{\theta}_9|$ respectively in (c), et al. This phenomenon indicates that there is a balanced relationship among the bearing measurement errors in the multi-station localization system, i.e., it is impossible for the system to improve the bearing measurement errors of all sensors when the sensors are in a general placement. When the sensors are in the optimal placement, the variation of the bearing measurement errors will be discussed in Section B.

(iii) Based on a comprehensive analysis of the above two phenomena, it can be seen that the bearing measurement error is closely related to the relative sensor-target geometry. For a single sensor, the absolute value of its bearing measurement error is directly proportional to the distance from the target. Therefore, there exists a balancing relationship of mutual compensation among the bearing measurement errors for the entire localization system. When the sensors are in the optimal placement relative to the target, the absolute values of bearing measurement errors are approximately constant values since the bearing measurement error of each sensor is 0 statistically. In addition, both the distances and bearings of each target relative to each sensor in Fig. 6 change simultaneously. According to the simulation results, the distance is the main factor affecting the bearing measurement error at this time.

2) HETEROGENEOUS SENSORS

In Fig. 7, the target is localized respectively using the two quantities of sensors in Table 3. The sampling intervals of Track 1 and Track 2 of S_2 are 0.5 km and 1 km respectively. In order to illustrate the relationship between the bearing

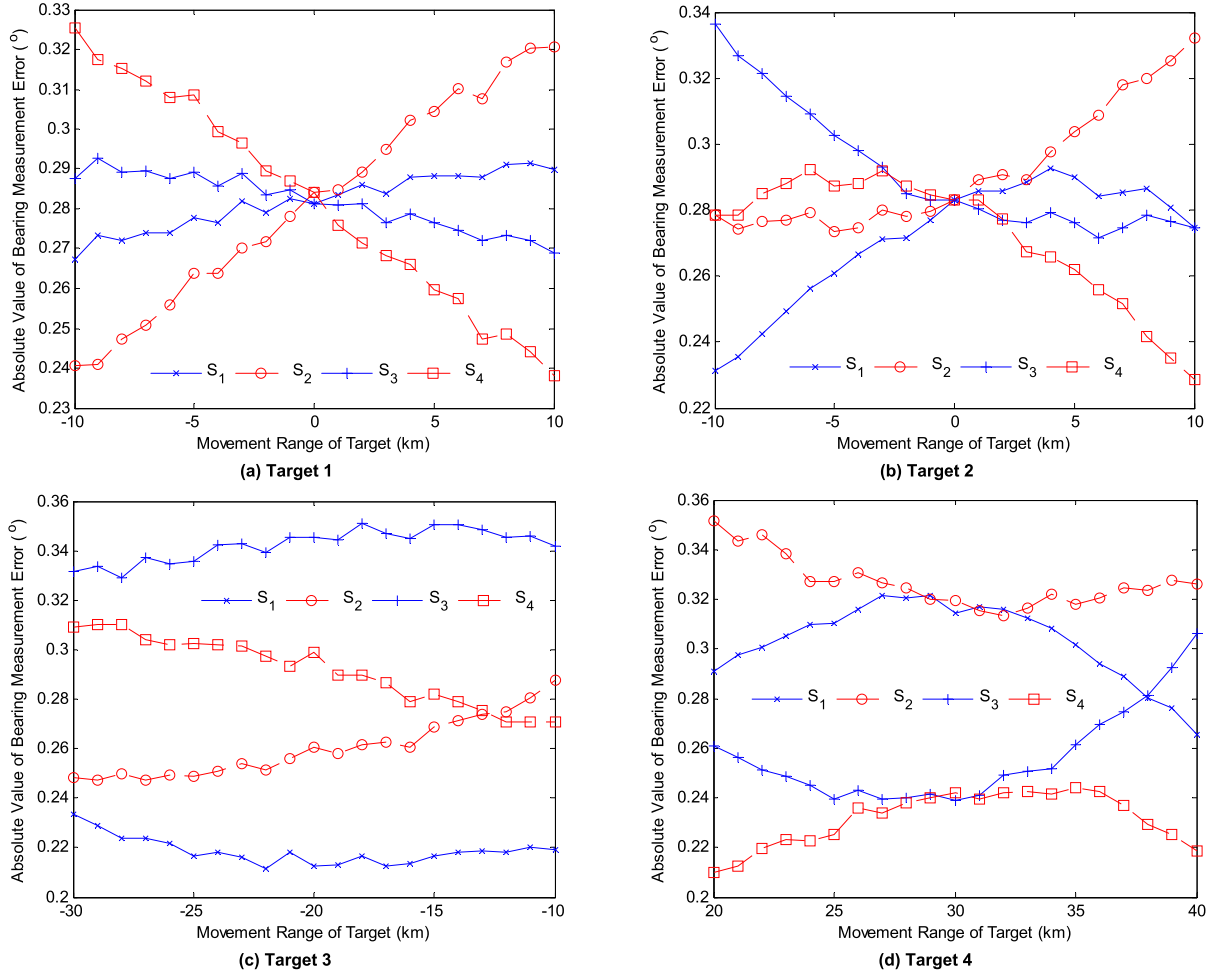


FIGURE 8. Absolute value of bearing measurement error when $n = 4$.

measurement error of heterogeneous sensor and the optimal placement form, the bearing measurement error of each heterogeneous sensor is compared with the corresponding homogeneous sensor with the same bearing measurement accuracy. For example, when $n = 3$ and S_2 moves along Track 1, the bearing measurement errors of S_1 - S_3 with bearing measurement accuracy of 1.5° , 1° and 0.8° are also simulated respectively while simulating the bearing measurement errors of S_1^* , S_2^* , S_3^* , so as to facilitate the comparison. Here only the simulation results of two situations are given: when $n = 3$ and S_2 moves along Track 1, and when $n = 4$ and S_2 moves along Track 2. The results are shown in Fig. 10 and 11. From them we know that:

(i) The bearing measurement error of each sensor is basically consistent with that of the homogeneous sensor with the same measurement accuracy. This result first verifies the correctness of (28) and (30), i.e., heterogeneous sensors can only obtain the optimal placement when they are “equidistant” from the target and the separation angles between adjacent sensors are equal; second, the bearing measurement error will achieve the optimal estimation when heterogeneous sensors present the optimal placement. At this time, the bearing measurement error of each sensor is 0 in a statistical sense.

(ii) As can be seen from Fig. 10(b), $|\Delta\hat{\theta}_2|$ is also smaller when S_2 is closer to the target. By comparing Fig. 10(a) and 10(c), it is known that since the bearings and distances of S_1^* and S_3^* relative to the target have not changed, the absolute values of their bearing measurement errors gradually increase for $|\Delta\hat{\theta}_2|$ gradually decreases. This phenomenon is also universal for heterogeneous sensor combining with Fig. 11, i.e., the bearing measurement error of each sensor also follows the symmetry principle when the sensors are in a general placement form.

(iii) As can be seen from Fig. 11, it has almost no impact on $|\Delta\hat{\theta}_4^*|$ when S_2 moves along Track 2, whereas the bearing measurement errors of S_5^* and S_6^* change significantly. This indicates that for a moving sensor, its impact on the bearing measurement error of adjacent sensors is greater than that on non-adjacent sensors.

B. BEARING MEASUREMENT ERROR UNDER OPTIMAL SENSOR PLACEMENT

This section mainly verifies that when the sensors are in the optimal placement form, the bearing measurement error of

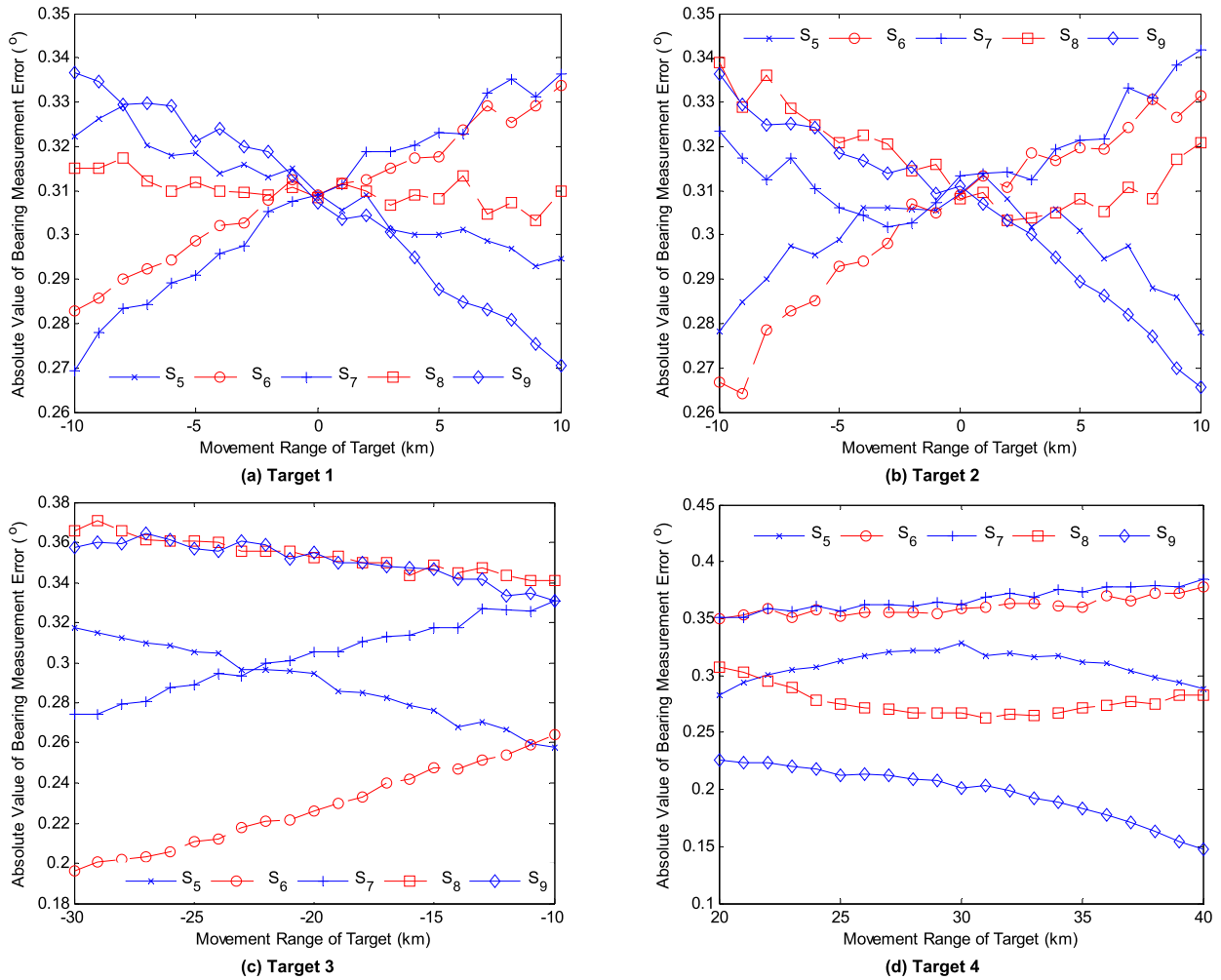


FIGURE 9. Absolute value of bearing measurement error when $n = 5$.

each sensor will remain at the optimal value and not change when the distances between the sensors and the target change.

Homogeneous sensors of the two quantities listed in Table 3 are used respectively to localize the target in Fig. 7, and the initial position of S_2 is at “○” place in the figure. The relative geometry of each sensor and the target remains unchanged, but the distance increases from 20 km to 40 km with a step size of 1 km. Heterogeneous sensors of the two quantities listed in Table 3 are used respectively to localize the target in Fig. 6, where the target is located at the coordinate origin. All sensors are moving towards the target direction, the horizontal axis movement range of S_2 is [30, 50] km with a step size of 1 km. The positions of the remaining sensors are calculated according to the principle of equal $d\sigma$, and the results are shown in Fig. 12 and 13. For convenience, the horizontal movement range of S_2 is used as the abscissa in Fig. 13. It can be seen from the figures that:

(i) When the sensors are in the optimal placement relative to the target, the bearing measurement error of each sensor remaining almost unchanged when changing the distances between the sensors and the target. $\Delta\hat{\theta}$ has reached the optimal state and $E[|\Delta\hat{\theta}|]$ is approximately a constant value

at this time according to the analysis of (45), therefore the estimation accuracy of $\Delta\hat{\theta}$ cannot be further improved when the distance is changed. Combining with Section A, it is known that $|\Delta\hat{\theta}_i|$ is proportional to d_i when the sensors are in a general placement form. However, it can be seen from this section that $\Delta\hat{\theta}$ remains at the optimal value and the above relationship no longer holds when the sensors are in the optimal placement. Therefore, the sensors must be in the optimal placement relative to the target in order to improve the bearing measurement errors of all sensors in the localization system.

(ii) Due to $E[|\Delta\theta|] \approx 0.8^\circ$ in Fig. 12, the results of (a) and (b) once again verify the consistency of the WLS estimation algorithm. The bearing measurement error is smaller with the higher measurement accuracy of sensor in Fig. 13, which is in line with the discussion about R in Section II.

C. RMS AND OPTIMAL MSE

The main verification in this section is: (1) MSE will obtain the optimal value when the sensors are in the optimal placement; (2) The localization accuracy of the target will be improved when $\tilde{\theta} - \Delta\hat{\theta}$ is used.

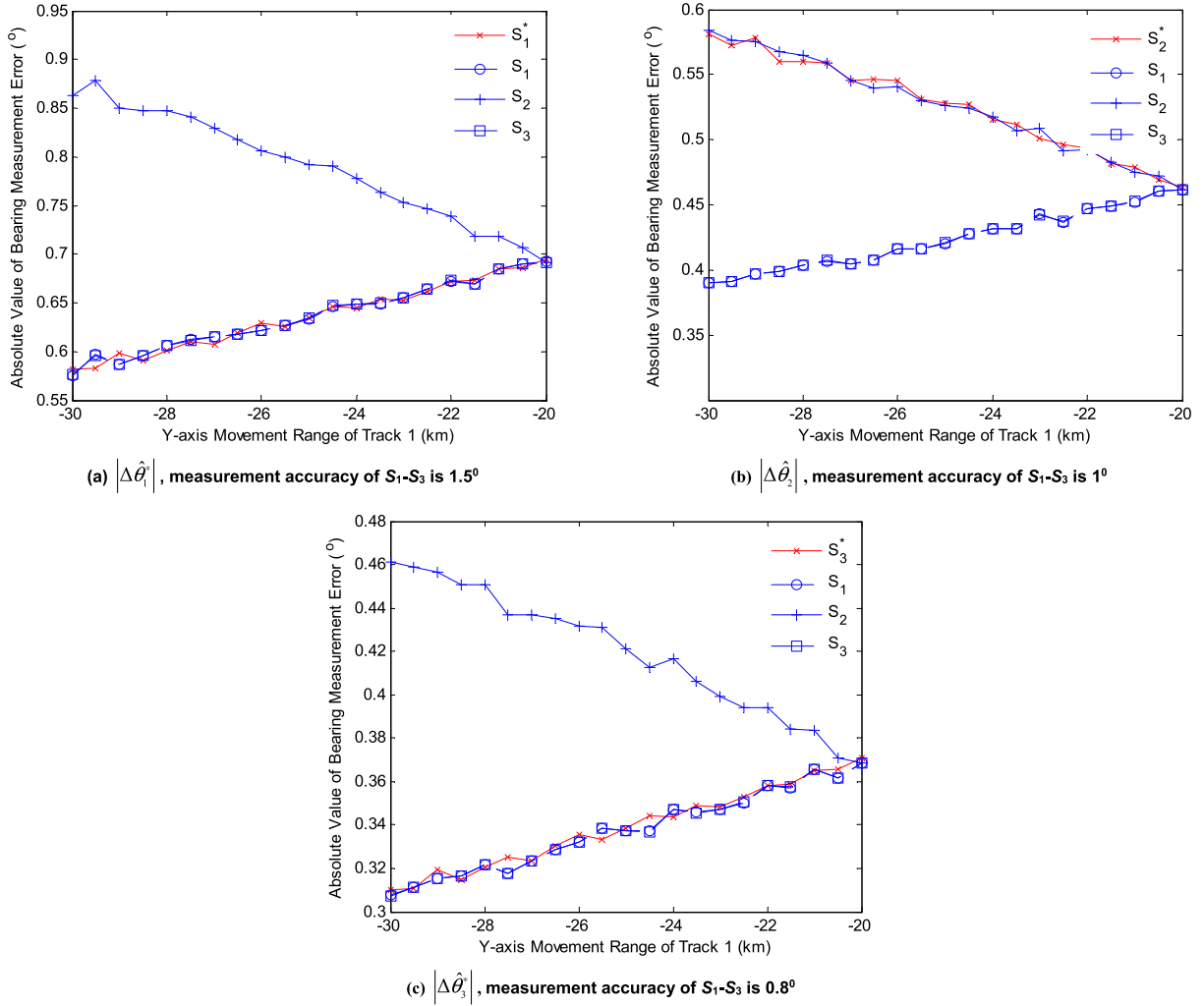


FIGURE 10. Absolute value of bearing measurement error when $n = 3$ and S_2 moves along Track 1.

In Fig. 7, the homogeneous/heterogeneous sensors in Table 3 are used to localize the target. When S_2 moving along track 1 and 2, the step size is 0.5km and 1km respectively. The results are shown in Fig. 14-17, where “modified” in the simulation results of RMS refers to the result when $\tilde{\theta} - \Delta \hat{\theta}$ is used for localization. As shown in the figures:

(i) By comparing the (a) and (b) graphs in Fig. 14 and 16 respectively, it can be known that homogeneous and heterogeneous sensors have the same MSE when the number of sensors is the same. This result is in line with the analysis of the optimal MSE in Section II, i.e., they will have the same MSE as long as the heterogeneous sensor has the same $d\sigma$ and bearing with the corresponding homogeneous sensor. MSE reaches the minimum value when the sensors are in the optimal placement form, which is consistent with Theorem 2. In addition, MSE is improved when the number of sensors is increased.

(ii) Homogeneous sensors and the corresponding heterogeneous ones almost have the same RMS when they have the same $d\sigma$ and bearing by comparing (a) and (b) graphs

in Fig. 15 and 17 respectively, which indicates that they basically have the same localization accuracy. RMS of “modified TLS” is significantly better than that of TLS algorithm, and the improvement degree when $n = 4$ is better than that when $n = 3$. This shows that since the bearing measurement errors of sensors approach the true values when using $\tilde{\theta} - \Delta \hat{\theta}$ for localization, the localization accuracy of the target can be significantly improved [39], [40].

D. EQ AREA

This section mainly conducts a simulation of the EQ area under the condition of the target traversing, and compares it with the true area value.

Homogeneous sensors with $n = 3$ are used to localize the target in Fig. 7, the measurement accuracy of each sensor is $\sigma = 0.5^\circ$. The range of the target on the horizontal and vertical axes is $[-25, 25]$ km and $[-15, 15]$ km respectively, with an interval of 1 km. After obtaining the estimated values of the bearing measurement errors, the relevant values of S_1, S_3 are used to calculate EQ area. In order to show the estimation effect of the WLS bearing measurement

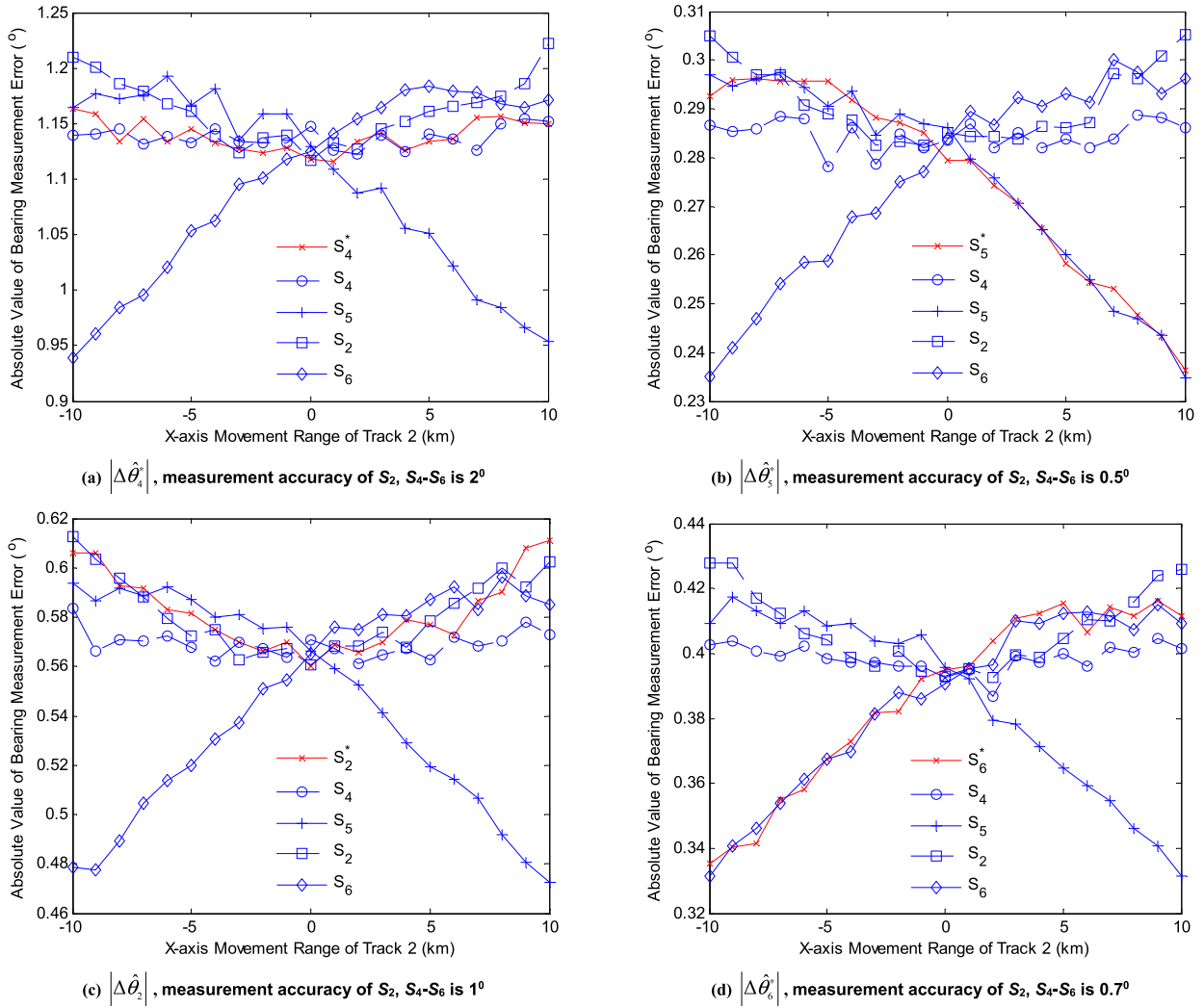


FIGURE 11. Absolute value of bearing measurement error when $n = 4$ and S_2 moves along Track 2.

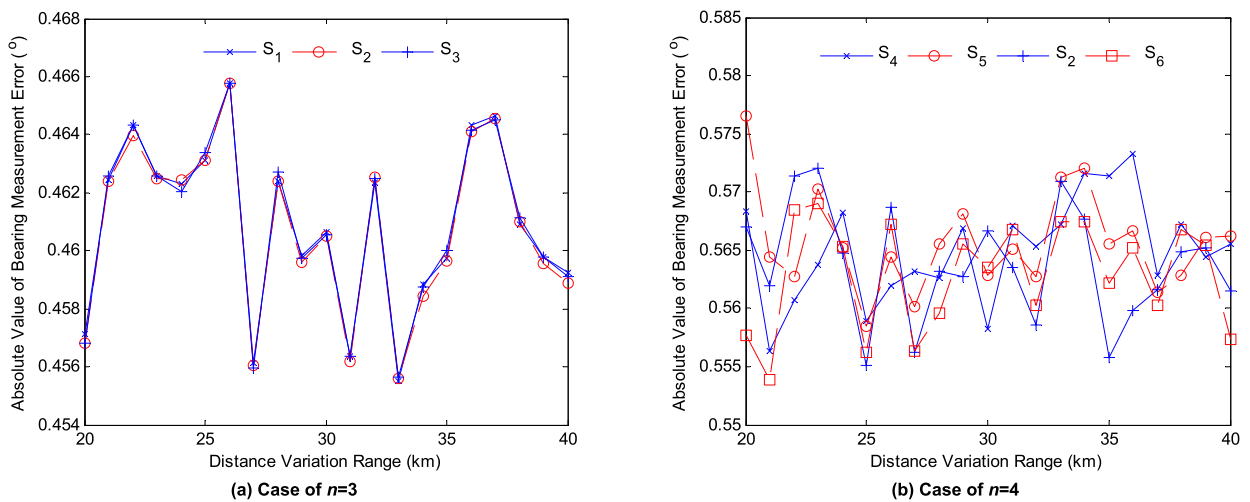


FIGURE 12. Absolute value of bearing measurement error when homogeneous sensors are in the optimal placement.

error algorithm, the estimated values and the true values of the bearing measurement errors are respectively substituted into (49), and the results are shown in Fig. 18 and 19.

It can be seen from the figures that there is a relatively large difference between the estimated value and the true value of the EQ area, but the distribution contours of them

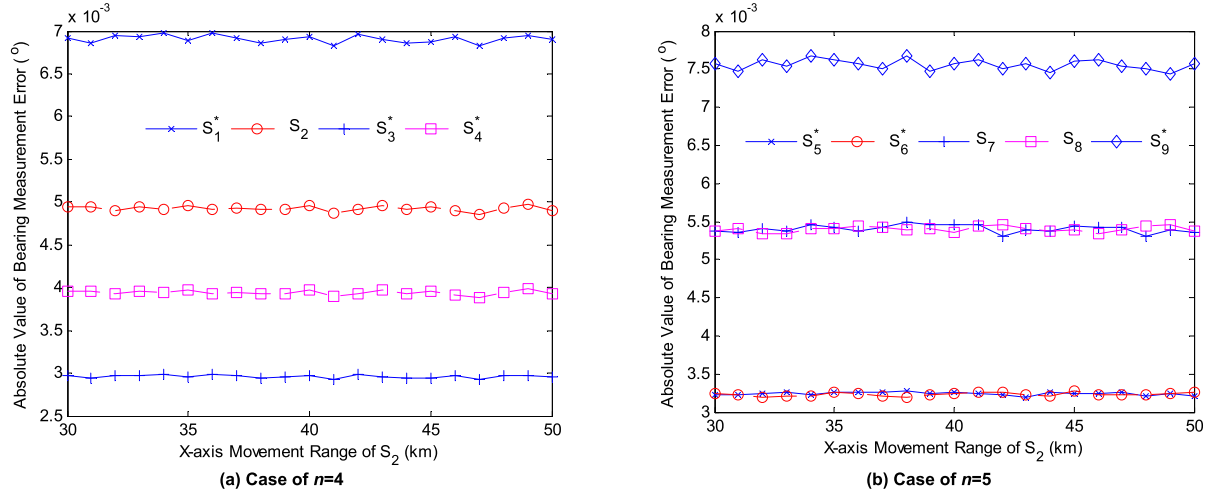


FIGURE 13. Absolute value of bearing measurement error when heterogeneous sensors are in the optimal placement.

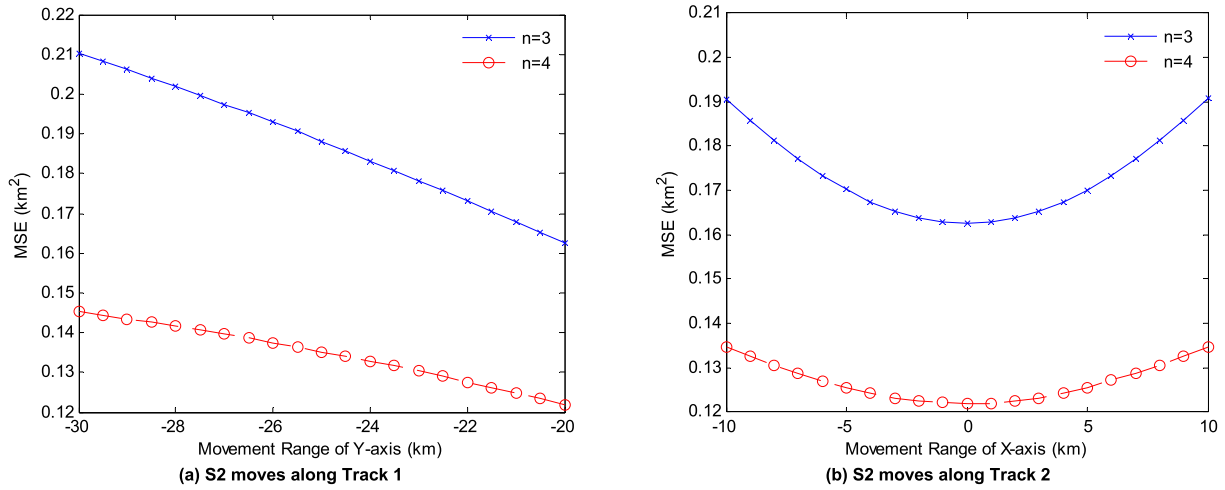


FIGURE 14. MSE for homogeneous sensors.

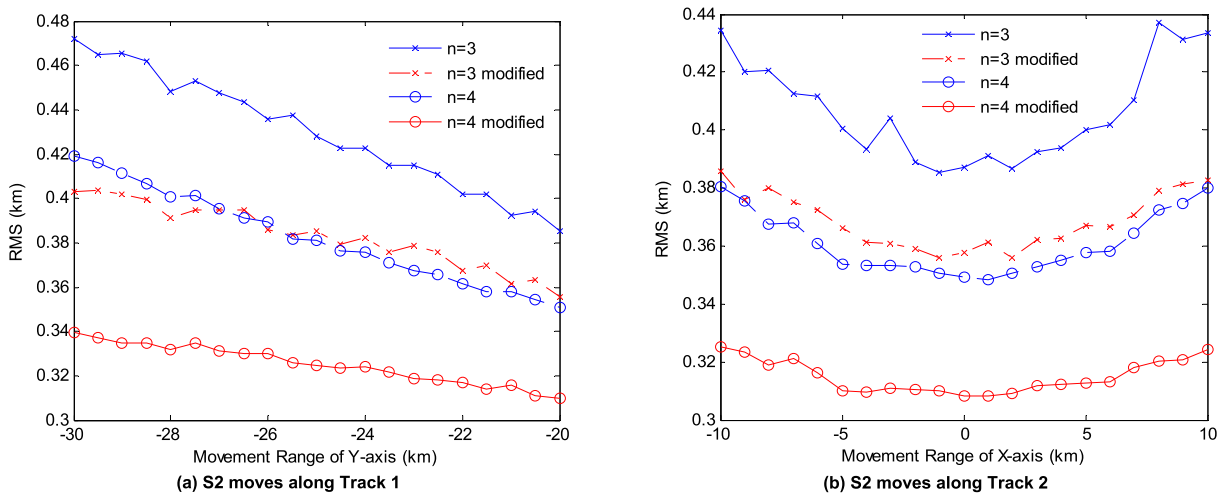


FIGURE 15. RMS for homogeneous sensors.

are basically the same. This results from the employment of minimal number of sensors in the WLS estimation algorithm. When the number of sensors increases, the estimated value

of the EQ area will approach the true value due to the estimation accuracy improvement of the bearing measurement error.

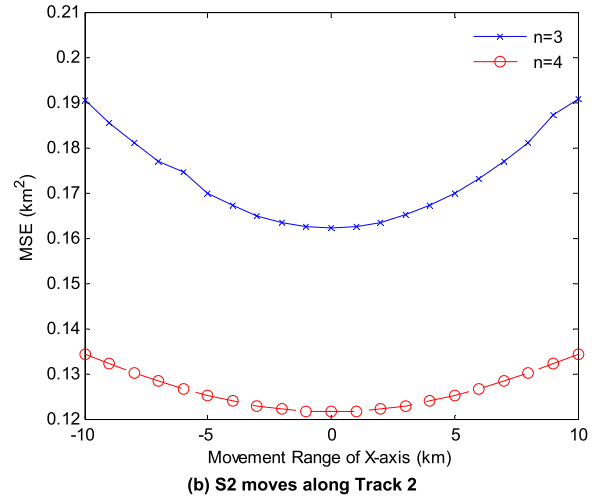
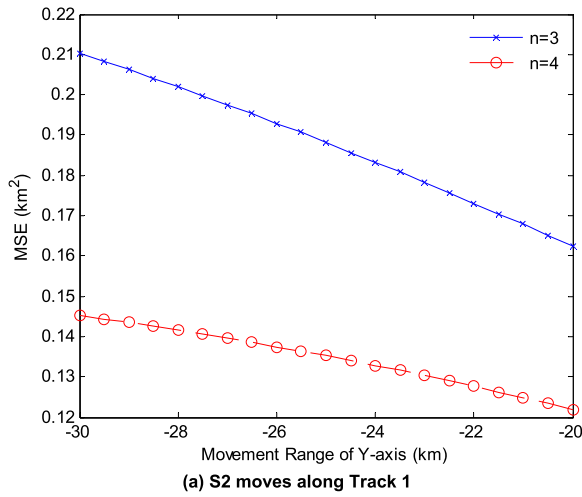


FIGURE 16. MSE for heterogeneous sensors.

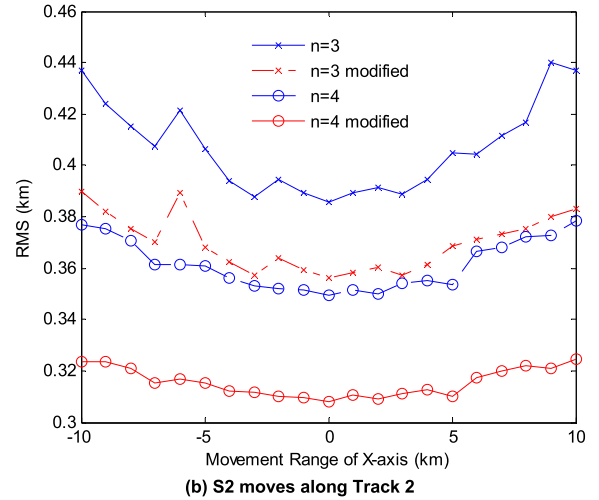
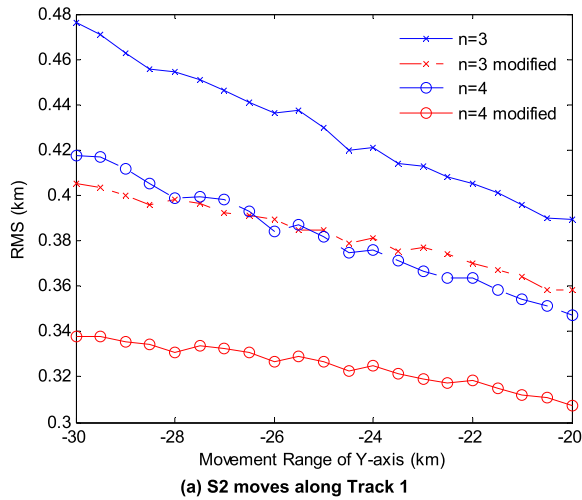


FIGURE 17. RMS for heterogeneous sensors.

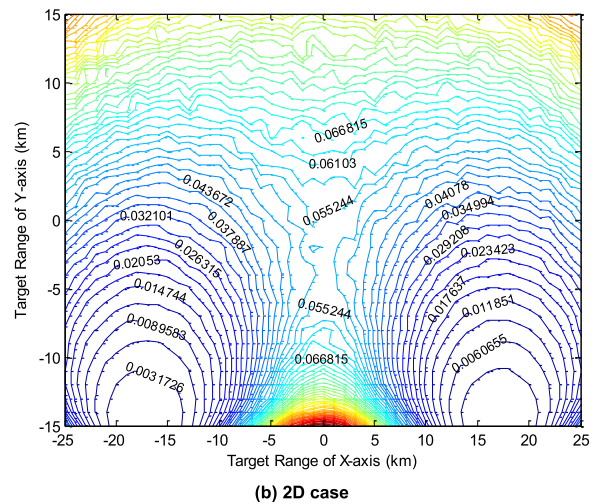
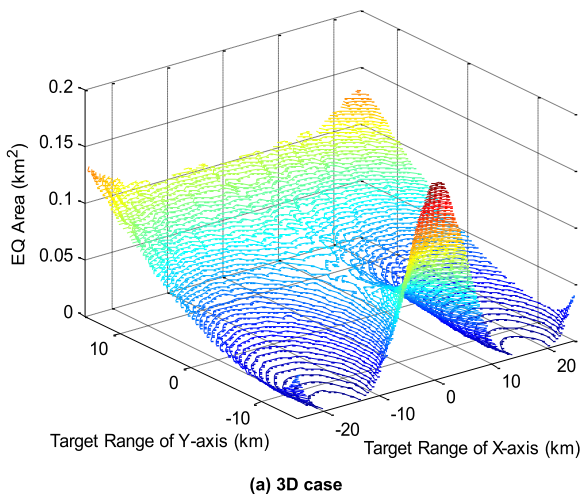


FIGURE 18. Estimated value of the EQ area.

VI. CONCLUSION

This paper first proposes a WLS algorithm for bearing measurement error estimation, with particular emphasis on

analyzing how to configure the sensors so that the bearing measurement error can be optimally estimated. Based on this analysis, we derive the optimal placements for both

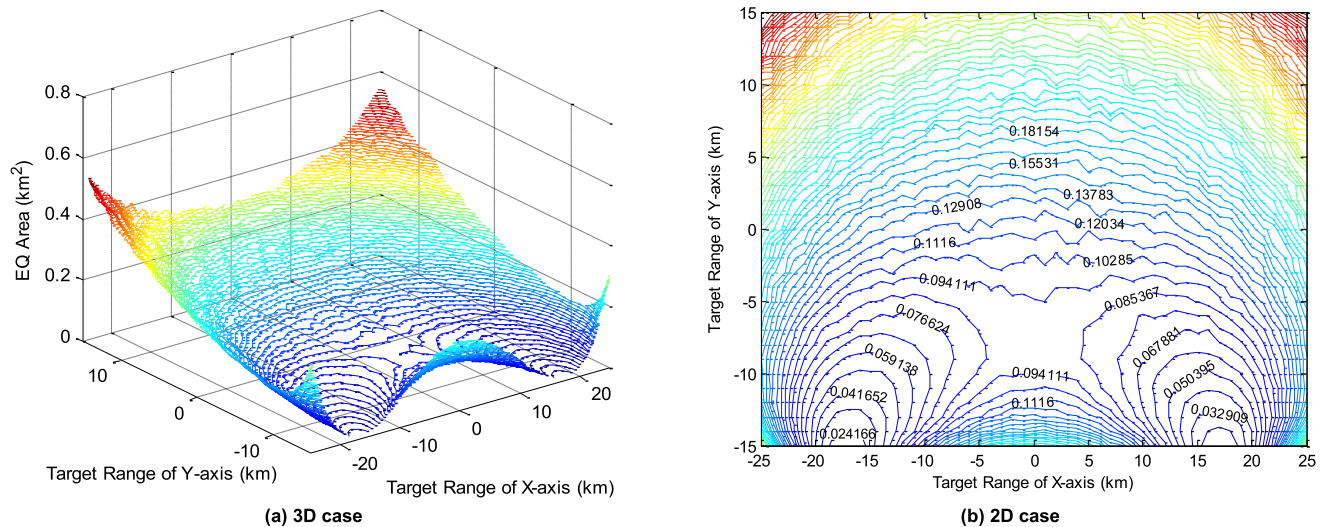


FIGURE 19. True value of the EQ area.

homogeneous and heterogeneous sensors and the interrelationships between different optimality criteria. Finally, the application of bearing measurement error estimation in EQ is investigated.

WLS bearing measurement error estimation algorithm is not complicated itself, but its significance lies in that it not only profoundly reveals the internal logic among the bearing measurement error, the error covariance matrix, and the localization accuracy, but also deepens the understanding of the necessary and sufficient conditions for the optimal localization accuracy. When the sensors are in a general placement, the bearing measurement errors of sensors satisfy the symmetry principle. This principle makes it impossible to obtain the optimal bearing measurement error estimation of each sensor regardless of how the sensors are configured. The closer the bearing measurement error is to the true value at this time, the smaller the values in the covariance matrix, and thus the better the localization accuracy. When the sensors are in the optimal placement form, the optimal localization accuracy can be obtained since the system satisfies both the relative sensor-target geometry condition and the measurement error condition described in the introduction at this time. Since the bearing measurement error of each sensor can be regarded as a “known quantity” at this time, there is no need to estimate it anymore. In other words, the localization accuracy will no longer be improved or the improvement will be insignificant when using the measurement correction values for localization at this time. The aforementioned analysis of the relationship between the bearing measurement error and the localization accuracy, as well as the correlation between the optimal bearing measurement error and the optimal placement form, is one of the key contributions of this paper.

Since (1) is a general model between the measured value and the measurement error, the above relationship between the bearing measurement error and the localization accuracy

has important enlightening significance and application value for other active/passive localization technologies. Although the measurement error is a random variable, its own change follows a specific pattern due to its internal relationship with the error covariance and the localization accuracy. Based on a deep understanding of this pattern, the measurement error can be estimated and thus the localization accuracy of the target can be effectively improved.

Existing research on the optimal placement form mainly focuses on homogeneous sensors, and points out that the bearing and distance constraint conditions that the sensors must satisfy are the necessary and sufficient conditions for the homogeneous sensors to obtain the optimal placement form. Through the research in this paper, it can be known that if the bearing measurement accuracy of sensor is regarded as the weight value of the distance, then heterogeneous sensors have the same optimal placement form as homogeneous sensors. The significance of this result lies in the substantial improvement of the cooperative localization ability of heterogeneous sensors, and it also has important enlightening significance and application value for other active/passive localization technologies. Therefore, the optimal placement form of heterogeneous sensors in passive BO localization is another key contribution of this paper.

Since this paper primarily investigates the optimal estimation of bearing measurement error, certain simulation phenomena were not analyzed in depth, such as why the absolute value of bearing measurement error follows the symmetry principle, and why a moving sensor exerts a greater influence on the bearing measurement error of adjacent sensor compared to a non-adjacent one. Obviously, these problems will become the focus of subsequent research. It should be noted that some simulation results in Section V are only applicable to the WLS estimation algorithm, their applicability to other algorithms requires further validation.

APPENDIX A

DERIVATION OF THE OPTIMAL MSE

In (40), let

$$D = \sum_{i=1}^{n-1} (n-i) \sin^2(i\Delta\theta) \quad (51)$$

two cases discussed respectively as follows.

A. TARGET LOCATED INSIDE THE POLYGON

When n is an odd number, (51) turns into

$$D = n \sum_{i=1}^{(n-1)/2} \sin^2\left(\frac{2\pi i}{n}\right) \quad (52)$$

due to [23]

$$\sum_{i=1}^{(n-1)/2} \sin^2\left(\frac{2\pi i}{n}\right) + \sum_{i=(n+1)/2}^{n-1} \sin^2\left(\frac{2\pi i}{n}\right) = \frac{n}{2} \quad (53)$$

let $k = n - i$, then obtain

$$\sum_{i=(n+1)/2}^{n-1} \sin^2\left(\frac{2\pi i}{n}\right) = \sum_{k=1}^{(n-1)/2} \sin^2\left(\frac{2\pi k}{n}\right) \quad (54)$$

thus in (53), then obtain

$$\sum_{i=1}^{(n-1)/2} \sin^2\left(\frac{2\pi i}{n}\right) = \frac{n}{4} \quad (55)$$

substitute it into (52), then obtain

$$D = n^2/4 \quad (56)$$

substitute it into (40), (29) can be obtained.

When n is an even number, (51) turns into

$$D = n \sum_{i=1}^{(n-2)/2} \sin^2\left(\frac{2\pi i}{n}\right) = n \sum_{i=1}^{n/2} \sin^2\left(\frac{2\pi i}{n}\right) \quad (57)$$

due to [23]

$$\sum_{i=1}^{n/2} \sin^2\left(\frac{2\pi i}{n}\right) + \sum_{i=(n/2)+1}^n \sin^2\left(\frac{2\pi i}{n}\right) = \frac{n}{2} \quad (58)$$

let $k = i - n/2$, then obtain

$$\sum_{i=(n/2)+1}^n \sin^2\left(\frac{2\pi i}{n}\right) = \sum_{k=1}^{n/2} \sin^2\left(\frac{2\pi k}{n}\right) \quad (59)$$

thus in (58), then obtain

$$\sum_{i=1}^{n/2} \sin^2\left(\frac{2\pi i}{n}\right) = \frac{n}{4} \quad (60)$$

substitute it into (57) and (40), (29) can also be obtained.

B. TARGET LOCATED OUTSIDE THE POLYGON

At this time, the denominator of (40) is (51) regardless of whether n is an odd number or an even number. Let $k = n - i$, then obtain

$$D = \sum_{k=1}^{n-1} k \sin^2\left(\frac{k\pi}{n}\right) = \sum_{k=1}^{n-1} \frac{k}{2} - \sum_{k=1}^{n-1} \frac{k}{2} \cos \frac{2k\pi}{n} \quad (61)$$

where

$$\sum_{k=1}^{n-1} \frac{k}{2} = \frac{n(n-1)}{4} \quad (62)$$

Let $w = e^{2\pi i/n}$, here i is the imaginary unit. Then obtain

$$\cos 2k\pi/n = (w^k + w^{-k})/2 \quad (63)$$

therefore

$$\sum_{k=1}^{n-1} k \cos \frac{2k\pi}{n} = \frac{1}{2} \left(\sum_{k=1}^{n-1} kw^k + \sum_{k=1}^{n-1} kw^{-k} \right) \triangleq \frac{s_1 + s_2}{2} \quad (64)$$

where

$$s_1 - ws_1 = w(1 - w^{n-1})/(1 - w) - (n-1) \quad (65)$$

after simplification, yields

$$s_1 = n/(w-1) \quad (66)$$

Let $a = w^{-1}$, we have

$$s_2 = \sum_{k=1}^{n-1} ka^k = \frac{n}{w^{-1}-1} \quad (67)$$

substitute (66) and (67) into (64), then obtain

$$\sum_{k=1}^{n-1} k \cos \frac{2k\pi}{n} = -\frac{n}{2} \quad (68)$$

(56) can be obtained combined with (62). Therefore, (29) can also be obtained when the target is located outside the polygon.

APPENDIX B

DERIVATION OF EQUATION (49)

Substitute (48a) and (48b) into (47a), then obtain

$$S_{\Delta S_2 AB} \triangleq L^2 \sin(2|\Delta\theta_2|)f_1/(2f_2) \quad (69)$$

where

$$f_1 = \sin^2(\tilde{\theta}_1 + |\Delta\theta_1|)$$

$$f_2 = \sin(\beta - |\Delta\theta_1| - |\Delta\theta_2|) \cdot \sin(\beta - |\Delta\theta_1| + |\Delta\theta_2|)$$

assume that $\Delta\theta_1$ and $\Delta\theta_2$ are small enough, then we have

$$f_1 \approx \sin^2 \tilde{\theta}_1 + |\Delta\theta_1| \sin 2\tilde{\theta}_1 \quad (70)$$

$$f_2 \approx \sin^2(\beta - |\Delta\theta_1|) \approx \sin^2 \beta - |\Delta\theta_1| \sin 2\beta \quad (71)$$

substitute the above two equations into (69), yields

$$S_{\Delta S_2 AB} \approx L^2 |\Delta\theta_2| \frac{\sin^2 \tilde{\theta}_1 + |\Delta\theta_1| \sin 2\tilde{\theta}_1}{\sin^2 \beta - |\Delta\theta_1| \sin 2\beta} \quad (72)$$

and similarly

$$S_{\Delta S_2 DC} \approx L^2 |\Delta\theta_2| \frac{\sin^2 \tilde{\theta}_1 - |\Delta\theta_1| \sin 2\tilde{\theta}_1}{\sin^2 \beta + |\Delta\theta_1| \sin 2\beta} \quad (73)$$

Substitute the above two equations into (46), then obtain

$$\begin{aligned} S_{EQ} &= \frac{L^2 |\Delta\theta_2| \sin \tilde{\theta}_1}{\sin \beta} \\ &\times \left(\frac{\sin \tilde{\theta}_1 + 2 |\Delta\theta_1| \cos \tilde{\theta}_1}{\sin \beta - 2 |\Delta\theta_1| \cos \beta} - \frac{\sin \tilde{\theta}_1 - 2 |\Delta\theta_1| \cos \tilde{\theta}_1}{\sin \beta + 2 |\Delta\theta_1| \cos \beta} \right) \\ &\triangleq \frac{L^2 |\Delta\theta_2| \sin \tilde{\theta}_1 f_3}{\sin \beta f_4} \end{aligned} \quad (74)$$

where

$$\begin{aligned} f_3 &= (\sin \tilde{\theta}_1 + 2 |\Delta\theta_1| \cos \tilde{\theta}_1) \cdot (\sin \beta + 2 |\Delta\theta_1| \cos \beta) \\ &\quad - (\sin \tilde{\theta}_1 - 2 |\Delta\theta_1| \cos \tilde{\theta}_1) \cdot (\sin \beta - 2 |\Delta\theta_1| \cos \beta) \\ &\approx 4 |\Delta\theta_1| \sin \tilde{\theta}_2 \end{aligned} \quad (75)$$

$$\begin{aligned} f_4 &= (\sin \beta - 2 |\Delta\theta_1| \cos \beta) \cdot (\sin \beta + 2 |\Delta\theta_1| \cos \beta) \\ &= \sin^2 \beta - 4 \cos^2 \beta (\Delta\theta_1)^2 \\ &\approx \sin^2 \beta \end{aligned} \quad (76)$$

substitute the above two equations into (74), and (49) can be obtained.

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BENCAI WANG received the M.S. degree from Zhengzhou Institute of Information Science and Technology, Zhengzhou, China, in 2007, and the Ph.D. degree from Naval Aviation University, Yantai, China, in 2011. He is currently a Lecturer with Wuhan Electronic Information Institute, Wuhan, China. His research interests include passive localization, information fusion, and optimized estimation.



ZHIYONG WANG was born in Wuhan, Hubei, China, in 1977. He received the B.S. and M.S. degrees in applied mathematics from the National University of Defense Technology, Changsha, China, in 1998 and 2003, respectively.

From 2004 to 2006, he was a Research Assistant with the Mathematical Modeling Laboratory. He is currently an Associate Professor with Wuhan Electronic Information Institute, Wuhan. He is the author of eight books and more than

40 articles. His research interests include applied mathematics and mathematics education.

Mr. Wang is a member of the Hubei Mathematical Society. He received several awards and honors, including the National Outstanding Teacher.



HONGTAO ZHAO was born in Hebei, China, in 1980. He received the Ph.D. degree in control science and engineering from the National University of Defense Technology, in 2011. He engaged in radar-related technology research and teaching work, including the application of artificial intelligence in radar systems, and has published more than ten academic articles in various journals.



XIAOTONG QI was born in Hubei, China, in 1992. She received the M.S. degree from Beijing Institute of Technology (BIT), Beijing, China, in 2018. She is currently a Faculty Member at a university, focusing her research on signal processing and information countermeasures, and she has published several articles in various journals.



ZHENG ZHANG was born in Hubei, China, in 1994. She received the master's degree in translation and interpreting from Central China Normal University, Wuhan, China. She participated in an academic program at Nanyang Technological University (NTU), in 2018. She is currently a Lecturer with Wuhan Electronic Information Institute, Wuhan. She has published several articles on pedagogy and translation and also contributed to the translation of two academic works.



LINGYAN ZHAO received the Master of Science degree in fundamental mathematics from Tianjin Normal University, Tianjin, China, in 2018. Her research focuses on theoretical analysis of ordinary differential equations and innovations in mathematics education methodologies. Her first-author publications include: "Generalization and Application of Rolle's Theorem" and "Estimation of Limit Cycle Numbers for a Class of Planar Piecewise Linear Hamiltonian Systems under Linear Perturbations." Her current research interest includes ordinary differential equations.

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